Degree Thesis

Master's level (Second cycle)

Comparing RAG and Fine-Tuned LLMs to ChatGPT for Domain-Specific Insights

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Abstract:

This thesis explores the use of retrieval-augmented generation (RAG) for enhancing large language models (LLMs) to answer domain-specific questions about Positive Energy Districts (PEDs). PEDs represent an innovative approach to achieving energy surplus and reducing greenhouse gas emissions, aligning with the European Union's goals for sustainable urban energy solutions. However, existing Question Answering (QA) frameworks lack the specialized design needed to tackle the complex challenges of PED projects. By constructing a structured corpus of academic literature about PEDs and fine-tuning models such as BERT and T5, the study evaluates their performance answering domain specific questions alongside GPT-enhanced pipelines and ChatGPT. Expert evaluations and cosine similarity analysis reveal that GPT-based pipelines outperform standalone models in accuracy, relevance, and readability. However, results also highlight limitations in semantic similarity as a sole metric for assessing response quality. This work highlights the role of structured domain-specific corpora and generative models in improving decision-making processes for stakeholders in urban sustainability. The findings suggest future research opportunities, including integrating real-time information retrieval and expanding knowledge bases to refine QA frameworks further. These contributions advance the implementation of energy-positive urban environments and support global efforts toward sustainable urban development.

Keywords:

Retrieval-Augmented Generation (RAG), Positive Energy Districts (PEDs), Large Language Models (LLMs), Fine-Tuning, Domain-Specific Corpus, BERT, T5, GPT-4, Cosine Similarity, Corpus Preprocessing, Knowledge Retrieval, Energy Efficiency, Stakeholder communication.

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1. Introduction

1.1 Background

Advancements in large language models (LLMs) have enabled artificial intelligence to address complex and specialized questions, offering new tools for addressing domain-specific challenges like those faced in the development of Positive Energy Districts (PEDs) (Yuan et al., 2024). PEDs are an emerging strategy in the European Union that represents a critical innovation for achieving energy transitions and reducing greenhouse gas emissions. However, the complexity of PED implementation which involves diverse stakeholders, integrated energy systems, and localized requirements poses significant challenges for effective information exchange and decision-making (Sassenou et al., 2024).

PEDs are urban areas achieving net zero CO2 emissions by generating more renewable energy than they consume, integrating buildings, energy systems, and transportation for sustainability and improved living standards (Urban Europe, n.d.). Aligned with the European Green Deal and the United Nations Sustainable Development Goals (SDGs), they utilize smart systems and local resources to combine energy-efficient buildings with sustainable transport like electric mobility. Renewable energy generation, storage, and distribution are integrated to provide collaboration among businesses, citizens, and governments (Vandevyvere, n.d.). The ability to efficiently retrieve and integrate information for PED stakeholders can significantly improve planning and decision-making processes, contributing to the broader goals of urban sustainability and energy transitions. By 2025, Europe aims to deploy 100 PEDs necessitating effective frameworks for managing their multi-stakeholder complexity (European Commission, n.d.).

Given the complex demands of PED implementation, solutions like question answering (QA) frameworks can play a pivotal role in streamlining decision-making and facilitating information exchange among stakeholders. QA frameworks interpret user queries and retrieve accurate, relevant answers by utilizing advanced NLP techniques and structured datasets (Van Otten, 2023). While extensively used in fields like healthcare and customer service, their adaptation to PED challenges, which are characterized by complex multi-stakeholder scenarios, remains underexplored and presents an opportunity for domain-specific innovation (Lämmerman, 2024).

This thesis investigates the potential of retrieval-augmented generation (RAG), a method that combines information retrieval with language generation (Li & Lai, 2024) to enhance QA frameworks for PED-related queries. By fine-tuning LLMs like BERT and T5 on a structured PED-specific corpus, this research evaluates their ability to deliver accurate and contextually relevant responses. Through expert evaluations and cosine similarity analysis the study examines the strengths and limitations of multiple LLMs, contributing to the development of intelligent systems that advance urban sustainability.

1.2 Research Gap, Aim and Questions

Despite the fact that QA frameworks have been successfully applied in domains like healthcare and customer service, their application to PED projects remains largely unexplored (Lämmerman, 2024). PEDs present unique challenges, requiring tools that can handle complex, multi-stakeholder scenarios and provide domain-specific answers to support informed decision-making. This gap limits stakeholders' ability to collaborate effectively in the planning and implementation of PEDs.

To address this need, this study aims to develop a domain-specific QA platform for PED-related inquiries, harnessing RAG to deepen the understanding of PED knowledge and improve communication among stakeholders. This thesis aims to design and evaluate such a platform, guided by the following research questions:

1. How can retrieval-augmented generation optimize the performance of large language models in answering domain-specific queries about Positive Energy Districts?
2. How does the performance of retrieval-augmented generation models compare to ChatGPT in answering domain-specific queries about Positive Energy Districts?

1.3 Structure of the thesis

This thesis is structured as follows: Chapter 1 introduces a brief background about PEDs, identifies the research gap, and presents the research questions. Chapter 2 reviews literature on PEDs, question-answering frameworks, fine-tuning of LLMs, and RAG processes. Chapter 3 outlines the methodology, including corpus development, model selection, and RAG implementation. Chapter 4 details the results, covering corpus creation, model fine-tuning, RAG evaluation, expert assessments, and cosine similarity analysis. Chapter 5 discusses the findings, emphasizing corpus construction, model fine-tuning, prompt refinement, and evaluation. Chapter 6 concludes with a summary of key findings, limitations, and future research directions.

2. Literature Review

This section is divided into four subsections. The first section examines PEDs and the second explores the use of chat-based question and answer platforms and Artificial Intelligence (AI) platforms to improve stakeholder communication, and decision making in PED projects. The final two sections review retrieval processes with a focus on RAG and fine-tuning of large language models.

2.1 Examining Positive Energy District Research

Although PEDs are a recent concept many have reached the planning phase. Zhang et al. (2024) provide a comprehensive analysis of 60 PEDs across Europe, highlighting their main features, challenges, and potential solutions. These urban areas are designed to generate renewable energy while maintaining net-zero greenhouse gas emissions, reflecting the growing emphasis on sustainable urban energy solutions. The objectives of PEDs align with the EU's energy strategy, supported by regulatory frameworks like Horizon Europe that promote research and innovation (Clerici Maestosi et al., 2024).

Key energy concepts in PEDs include demand, consumption, production, supply, balance, and performance. Energy demand refers to the amount required by districts; consumption indicates how much is used; production involves generating energy; supply pertains to delivering energy; balance reflects the relationship between consumption and production; and performance measures the difference between imported and exported energy (Sassenou et al., 2024). PEDs are designed to be energy-efficient and flexible within urban networks of interconnected buildings. They integrate various systems and infrastructures, ensuring interaction between buildings, users, and regional energy, mobility, and ICT [Information Communication Technology] systems. Their goal is to secure energy supply while promoting social, economic, and environmental sustainability (Vandevyvere, n.d).

Despite the potential benefits of PEDs, their implementation faces challenges due to the diversity of stakeholders and the complexity of integrating multiple energy systems at the district level. These challenges are intensified by a lack of practical experience, limited knowledge of PED operations, and socio-cultural and historical differences. While PEDs are a European initiative, similar efforts are emerging globally, opening opportunities for international collaboration and knowledge sharing. As PEDs continue to evolve, they represent a viable approach toward sustainable, low-carbon urban environments, but their success depends on overcoming these early-stage challenges (Sassenou et al., 2024).

2.2 Examining the Research on Question Answering Frameworks

A QA framework is a system designed to automatically respond to questions posed in natural language by humans. These systems can retrieve answers from structured databases (knowledge bases) or unstructured natural language documents like webpages. QA frameworks can be categorized by the types of questions they handle such as simple factual queries, requests for definitions or explanations, and inquiries about hypothetical situations (Xue et al., 2024). QA frameworks often utilize complex architectures including retriever-reader systems or transformer-based models to process large collections of text or trained data for answering questions. They can operate within different subject matter domains: closed-domain systems are designed for specific areas, while open-domain systems handle questions on any topic and rely on general world knowledge or large-scale datasets (Fu et al., 2023).

QA systems are crucial to enhancing information retrieval in natural language processing. These systems undergo several stages to ensure effective performance. The question analysis phase establishes a foundation through query formulation and question classification utilizing linguistic techniques like parsing and synonym analysis to address term mismatches. Document retrieval identifies relevant documents using information retrieval methods that assess the relevance of terms and documents. In the answer extraction stage, generative methods formulate answers in natural language improving user-friendliness. Finally, post-processing enhances answer quality by filtering documents and validating responses. Together, these stages enable QA frameworks to process large volumes of unstructured data and provide accurate answers significantly enhancing user experience (Zhu et al., 2021).

User queries are central to QA frameworks as they reflect a user's intent to retrieve specific information from a knowledge base or community platform. These queries can take various forms including factual inquiries, subjective requests for opinions, or social questions aimed at facilitating discussions (Chen et al., 2012). To effectively process user queries, several preprocessing stages are involved. The first stage, text preprocessing, standardizes input by parsing it into tokens, which are the smallest units of text such as words, subwords, or symbols used for analysis. It also involves lowercasing text, removing stopwords such as common words like "and" or "the" that add little meaning, and eliminating unnecessary punctuation (Zhu & Luo, 2023). This standardization is crucial for ensuring that the input is clean and comprehensible for subsequent processing. Following this, natural language understanding (NLU) techniques such as Named Entity Recognition (NER) detect important entities like names and locations. Intent recognition helps clarify user goals, while relation extraction identifies connections between entities. This comprehensive understanding enables the system to decide whether to retrieve information from a knowledge base or generate a response using a language model (Humans of Nesh, 2021). By incorporating these elements, QA systems can enhance their ability to deliver accurate and relevant responses, ultimately improving user experience and satisfaction

2.3 Techniques in Document Retrieval

The document retrieval stage involves searching for relevant documents or passages using generated search queries. This process uses information retrieval (IR) techniques to measure the importance of terms, such as Term Frequency-Inverse Document Frequency (TF-IDF), and ranking algorithms like BM25 to prioritize results. BM25 enhances relevance by balancing term frequency and document length, while methods inspired by web search engines, such as those used by Google and Bing, are also applied to improve retrieval effectiveness (Zhu et al., 2021).

Traditional methods like TF-IDF assess the significance of a document by evaluating how often terms appear within it, while also considering their distribution across the entire corpus. This allows the system to prioritize content that is more relevant to user queries. Similarly, BM25 calculates relevance scores by considering term frequency, inverse document frequency, and document length normalization, resulting in improved performance across various retrieval tasks (Prasan, 2024). However, these techniques often struggle with complex queries that require more contextual awareness (Kadhim, 2019). To address this limitation, RAG emerges as a hybrid approach that combines the strengths of retrieval and generation models, enabling systems to access both up-to-date knowledge and contextually relevant data.

The RAG system operates in two phases: retrieval, which extracts relevant data from external sources, and generation, which uses this information to produce contextually accurate outputs that closely align with user queries (Figure 1). Central to the generation phase are models like GPT-4 which use retrieved data to produce detailed and accurate responses thereby enhancing the reliability and relevance of outputs in applications requiring contextual awareness (Yu et al., 2024). RAG is an AI framework that enhances LLMs by integrating external knowledge. This approach mitigates the limitations of traditional LLMs, which often produce outdated or inaccurate information due to reliance on pre-existing training data and a tendency to hallucinate facts (Martineau, 2023).

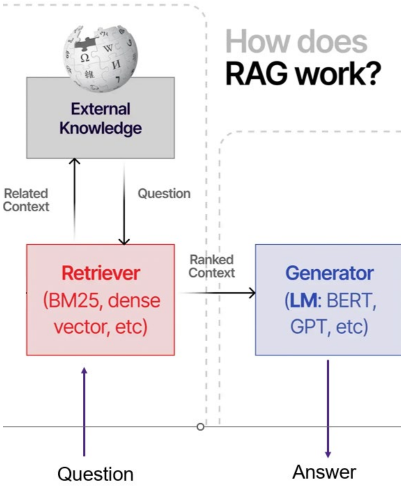

Figure 1: Overview of Retrieval-Augmented Generation (RAG) Workflow

A key advantage of integrating RAG with LLMs is RAG's ability to provide real-time, dynamic, and contextually informed responses across sectors such as healthcare and law, thereby enhancing user experience and operational efficiency. For example, RAG-based chatbots in customer service can utilize real-time data for relevant answers, while content recommendation systems analyze user behavior for personalized suggestions (Regunath, 2023). Central to RAG's effectiveness are embedding techniques that improve semantic understanding of queries and documents. Embeddings are numerical vector representations, where each query or document chunk is converted into a mathematical format capturing its meaning. The system then computes cosine similarity, a measure of how similar two vectors are based on the angle between them. A higher similarity score indicates greater semantic closeness, helping identify documents most relevant to the query (Kimura et al., 2024). Various embedding models, including sparse encoders like BM25 and dense retrievers built on architectures such as BERT, play crucial roles in this process (Gao et al., 2024).

Different embedding techniques further enhance RAG's retrieval capabilities. For instance, Word2Vec employs neural networks to generate word embeddings that capture semantic relationships, while GloVe (Global Vectors for Word Representation) uses word co-occurrence statistics to build embeddings reflecting global semantics (Mars, 2022). OpenAI's text embedding-ada-002 is a cutting-edge model that generates dimensional embeddings with deep semantic understanding, excelling in applications including natural language understanding and multilingual tasks (Ghosh, 2024). Similarly, SentenceTransformers like Sentence-BERT (SBERT) and its variants, such as all-MiniLM-L6-v2, excel in encoding sentence-level semantics into dense vector representations, making them highly effective for semantic search and large-scale similarity comparisons (Dhini et al., 2023).

Despite its strengths, RAG faces challenges related to complexity, data privacy, and information quality. The integration of retrieval and generation components increases computational demands, and reliance on external data sources raises security concerns. Moreover, the quality of the final output heavily depends on the success of the retrieval phase (Angeln, 2024). Addressing these challenges through careful design and robust security measures is crucial for maximizing RAG's effectiveness in real-world applications.

2.4 Model Fine-tuning for specific domains

Fine-tuning is a vital process that enhances pre-trained models by training them on smaller, domain-specific datasets. This method builds on existing knowledge, improving performance on specialized tasks while requiring less data and computational resources (Amanatullah, 2023). The goal is to retain the original capabilities of the pre-trained model while adapting it for specific uses (Craig, 2024). Fine-tuning can be approached in three main ways: Unsupervised Fine-Tuning, which uses unlabeled text from the target domain to improve language understanding; Supervised Fine-Tuning, which employs labeled data for targeted learning; and Instruction Fine-Tuning, which provides natural language instructions to enhance task execution (Balavadhani Parthasarathy et al., 2024).

According to Craig (2024), the fine-tuning process for an LLM involves several key stages. It begins with dataset preparation, where a task-specific dataset is cleaned and formatted into input-output pairs to guide the model's behavior, such as for instruction tuning or sentiment analysis. Next, model initialization sets initial parameters to ensure optimal performance and prevent issues disrupting training. The training environment setup stage configures the infrastructure, selects relevant data, and defines the model architecture and hyperparameters for fine-tuning. During partial or full fine-tuning, the model’s parameters are updated using the task-specific dataset; full fine-tuning adjusts all parameters, while partial fine-tuning targets specific layers for efficiency. Evaluation and validation assess performance on unseen data using metrics to ensure generalization and detect overfitting. Finally, the model is deployed and integrated into the system for tasks like text generation or query understanding.

According to Mudadla (2023) fine-tuning offers key advantages, such as applying transfer learning to boost performance on limited data, faster training by building on pre-trained models, effective domain adaptation, and reduced data requirements. However, it also has drawbacks, including the risk of overfitting on small datasets, task mismatch if the pre-trained model isn't well-aligned, the loss of learned features, as well as a dependency on the quality of the pre-trained model.

3. Methodology

This methodology outlines the process of evaluating how LLMs can be adapted to answer domain-specific questions about PEDs. Our approach involves key steps such as corpus creation, data extraction, preprocessing, model fine-tuning, and the integration of RAG. The goal is to assess whether an LLM utilizing RAG can provide accurate and contextually relevant answers to both fact-based and open-ended queries related to PEDs.

To achieve this, three LLMs were employed: BERT, T5, and GPT-4 alongside its optimized variant, GPT-4o. Two distinct corpora were created: the **fine-tuning corpus**, used to fine-tune BERT and T5, and the **RAG corpus**, used as a source of information for answering questions during RAG implementation.

General, specific, and synthetic (fake) questions about PEDs were posed to the models, and their outputs were compared to responses generated by ChatGPT, an open-domain LLM, to evaluate performance differences. The answers were evaluated using a combination of expert review and cosine similarity for comparing specific answers in the corpus.

The following sections detail the steps of data collection, preprocessing, choice of language models, model fine-tuning, RAG integration, and output evaluation, illustrating how these components collectively address the research objectives.

3.1 Corpus Development: Data Collection and Preparation

The methodology begins with the careful selection of academic literature that forms the foundation of the corpus. This step ensures that the language models are fine-tuned on accurate, reliable, and domain-specific content. Selection criteria included peer-reviewed articles from SCOPUS explicitly containing "Positive Energy District(s)" in their titles and published between 2022 and 2024, a timeframe chosen to capture the latest advancements in PED research. This workflow is summarized in Figure 2, highlighting the steps from article selection to corpus structuring.

This process resulted in 50 articles being downloaded in batches as PDFs. By manually curating this specialized corpus, we ensured consistent and reliable access to PED-related content tailored to domain-specific queries, addressing potential gaps in existing open-domain LLMs. With this literature in place, the focus shifted to preparing a structured corpus through a combination of automated tools and manual review

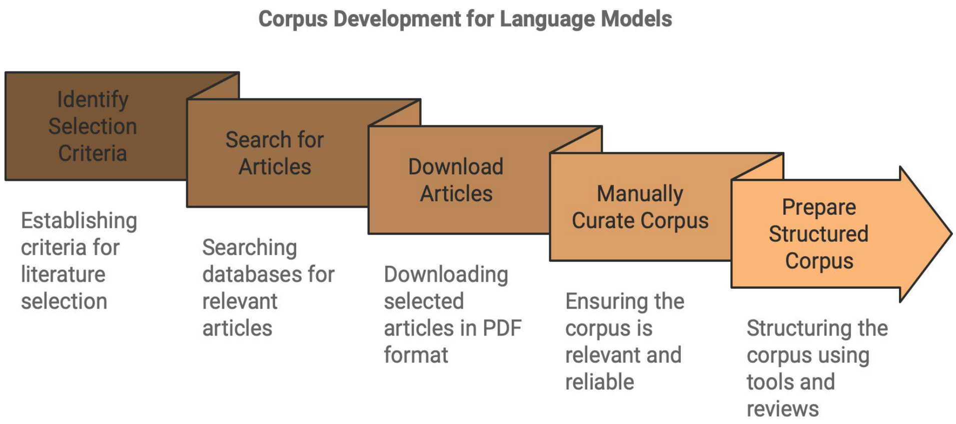

Figure 2: Corpus Development Process for Language Models
This diagram summarizes the steps for creating a corpus, including identifying criteria, searching for articles, downloading, manual curation, and preparing a structured format.

3.1.1 Data Extraction and Preprocessing

Extracting structured data from the downloaded documents was the first step in preparing the corpus for fine-tuning and retrieval tasks. This process combined automated tools with manual review to ensure accuracy and comprehensiveness.

Manual data extraction was performed on two articles containing pre-existing question-and-answer pairs specific to PEDs. Forty questions and their corresponding answers were directly copied from these documents and stored in a comma-separated values (CSV) file (see Appendix E). This supplementary dataset provided high-quality examples of questions and answers for the RAG process, enabling the models to generate accurate and contextually relevant answers to PED-related queries.

For automated extraction, Grobid (Generation of Bibliographic Data) was selected for its proven ability to parse and structure academic documents effectively (Astudillo, 2024). Operating on a local machine via Docker, Grobid converted downloaded PDFs into XML (Extensible Markup Language) format, a machine-readable representation of document content. XML was chosen because its hierarchical structure organizes text into clearly defined elements using tags (e.g., <abstract>, <div>, <head>, <p>). This organization enabled precise identification of key sections, such as abstracts, methods, and conclusions, which were critical for the RAG implementation.

To prepare the extracted data for fine-tuning the language models, Python scripts were developed to process the XML files. A primary challenge was handling the varied structures of XML documents. The scripts relied on tags like <div> to group related content, <head> tags to identify section headings, and <p> tags for corresponding text. The n attribute within

<head> tags, which encodes section numbers (e.g., "1", "1.1"), was used to preserve the logical hierarchy of the original documents. For cases where tags or attributes were missing, placeholder names such as "Unnamed Section" were assigned to maintain completeness.

The parsed sections, subsections, and their corresponding text were then compiled into a single structured JSON file. JSON (JavaScript Object Notation) was chosen for its compatibility with Python-based processing and its ability to represent hierarchical data in a clear and accessible format. Its key-value pair structure, where data is organized as unique identifiers (keys) and their associated information (values), made it straightforward to preserve the organization of the XML data while ensuring ease of use in downstream tasks, such as retrieval and fine-tuning. Additionally, a separate script processed question-answer pairs from the CSV file into JSON format for fine-tuning. By converting the data into structured JSON, the corpus became easily accessible to machine learning models, enabling precise and efficient use of the information during retrieval and generation tasks.

Once the corpus was organized into a structured JSON format as seen in Figure 3, additional preprocessing steps were applied to standardize the data. These steps included converting all text to lowercase to ensure uniformity, removing stop words to reduce noise, and eliminating extraneous formatting, such as special characters or unnecessary spaces. Stop words, such as "the" or "and" were removed because they carried little semantic meaning and could hinder the model's ability to identify relevant patterns.

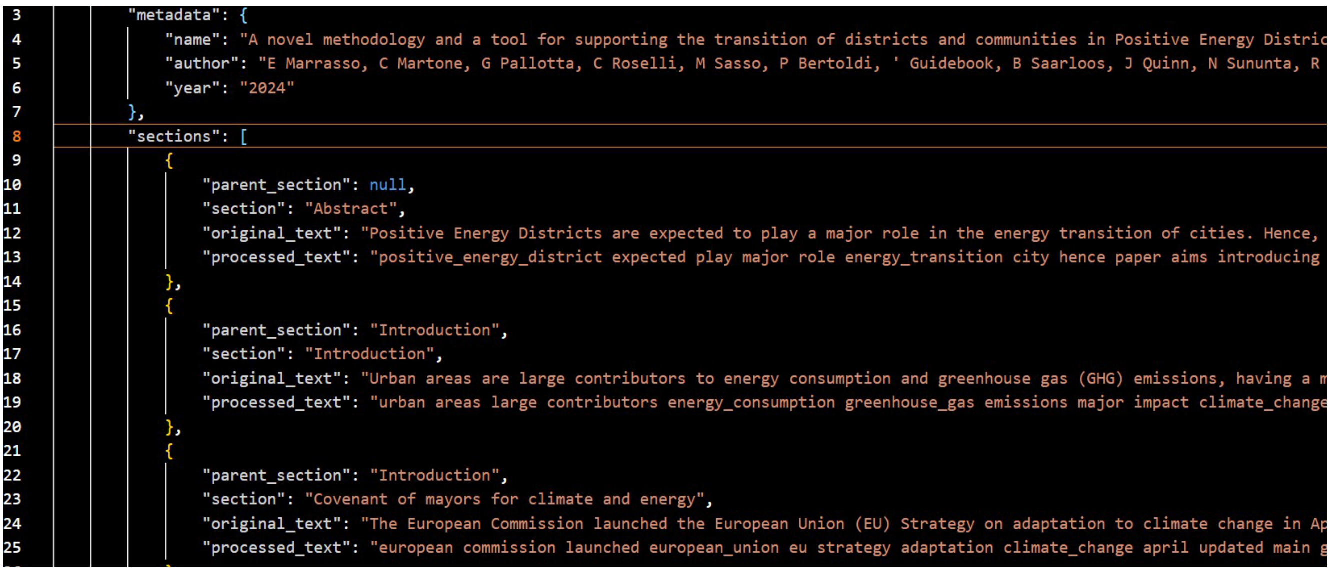

Figure 3: The Structured Corpus Before Chunking
This screenshot displays the hierarchical structure of the RAG corpus before chunking. Each section is represented with metadata, including the parent section, section title, original text, and processed text. This organization ensures traceability and preserves contextual relationships, forming the foundation for subsequent preprocessing and embedding.

Specific terms related to PEDs were also standardized to ensure consistency. For instance, "districts" was normalized to "district." Additionally, tokenization was applied to preserve domain-specific terminology, such as "positive energy district" (seen on line 13 in Figure 3), "climate change," and "renewable energy." Tokenization refers to the process of breaking text into smaller units, such as words, subwords, or phrases. In some cases, it also involves combining words that commonly appear together to preserve their contextual meaning for language models. These preprocessing steps minimize inconsistencies and noise, optimizing the corpus for fine-tuning the language model and improving its ability to generate precise, contextually relevant answers.

3.1.2 Corpus Refinement: Handling Issues and Final Preparation

The refinement of the corpus involved iterative testing and addressing specific issues identified during development and preparation. Early in the process, smaller trial corpora were created to serve as controlled datasets for debugging data extraction and preprocessing workflows. These corpora were particularly useful for evaluating tokenization and normalization strategies. Insights from this testing phase informed the design of the final preprocessing pipeline, which addressed three critical issues in the RAG corpus: irrelevant content, incomplete data extraction and token length limitations. Resolving these issues was essential to improving the general performance of the models.

The first issue involved one of the selected academic articles, which contained extensive references to ChatGPT. These references influenced the initial model outputs in undesirable ways, introducing noise and reducing the relevance of responses to PED-related queries. To mitigate this, we manually removed the article and all its associated content from the corpus.

The second issue arose when specific questions failed to retrieve content that was known to exist in the corpus. A manual evaluation revealed that the extraction process from the XML files to the structured JSON format had not worked as intended for certain articles. In these cases, only the abstract or partial sections were extracted, resulting in incomplete entries in the corpus. Upon reviewing the XML code for these articles, we discovered inconsistencies in their structure that the original extraction script could not handle. To address this, the XML extraction script was revised to dynamically accommodate diverse XML structures and processed nested tags more robustly. This revision increased the size of the corpus by 63%. Following this improvement, data cleaning, normalization, and tokenization steps were reapplied to the expanded corpus, which is referred to as the Fine-Tuning Corpus.

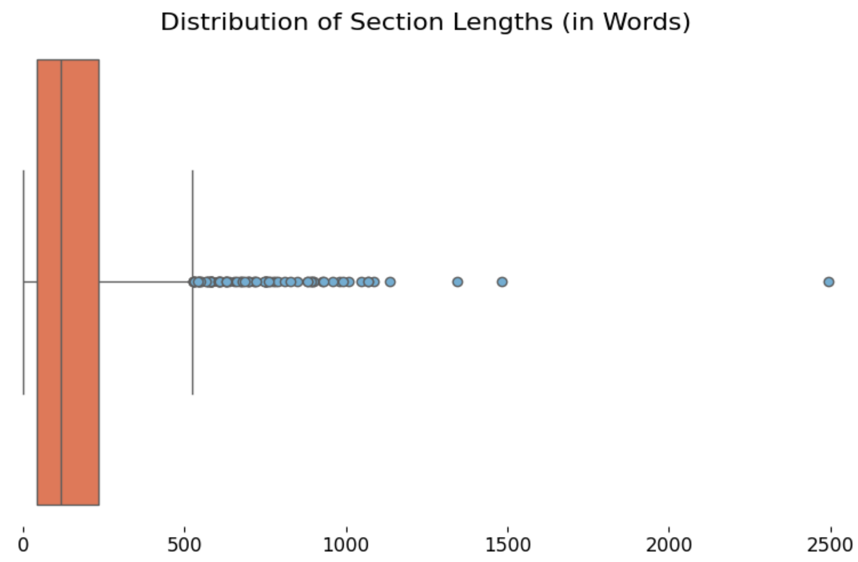

Figure 4: Distribution of Section Lengths (in Words)
This boxplot visualizes the distribution of section lengths within the dataset, measured in words. The majority of sections cluster tightly around shorter lengths, as indicated by the dense grouping near the lower end of the scale. 80 sections extend beyond the interquartile range, revealing the presence of outliers with substantially higher word counts.

The final issue was related to the length of certain sections within the corpus. During a comparison of retrieved sections with a manual review of the expanded corpus, we found that 80 sections exceeded the 512-token input limit of the chosen language models, as illustrated in the distribution of section lengths seen in Figure 4. To resolve this, we developed a Python script to implement a chunking mechanism after preprocessing and data cleaning were completed. This script systematically divided all sections into smaller pieces called "chunks" while preserving metadata, such as section names, in dictionary format. Retaining metadata ensured that hierarchical relationships were maintained, enabling the models to associate chunks with their original sections during encoding for RAG. This segmentation was essential for preventing the removal of important information and improving the efficiency and accuracy of the retrieval process. The resulting segmented corpus is referred to as the RAG Corpus.

By addressing these issues and refining the corpus, we ensured that the models could utilize both the fine-tuning and RAG corpora effectively. These adjustments laid the groundwork for model fine-tuning and retrieval processes which are explained below.

3.2 Model selection and development

While our initial intention was to evaluate a single language model (BERT) in comparison to Chat GPT-4o, limitations in BERT's outputs prompted the exploration of alternative models, including T5 and GPT-4. Each model's strengths and shortcomings informed improvements to our RAG pipeline, resulting in a comparative evaluation of their effectiveness in generating domain-specific responses.

3.2.1 Fine-Tuning BERT for QA Using the Fine-Tuning Corpus

BERT (Bidirectional Encoder Representations from Transformers) was chosen as our initial model due to its status as a pre-trained LLM that generates deep bidirectional representations of text. This means it analyzes each word in a sentence by simultaneously considering both the words that precede it and those that follow, enabling a nuanced understanding of word meanings and sentence structures (Devlin et al., 2019). BERT's architecture is based on the transformer model which uses self-attention mechanisms to weigh the significance of words relative to one another, producing contextualized word embeddings sensitive to their specific contexts (Rogers et al., 2021). Pre-trained on large text corpora such as Wikipedia and BookCorpus, BERT captures rich contextual relationships in text, making it particularly well-suited for our task of question answering (QA).

To adapt BERT for QA, we fine-tuned the model using the Fine-Tuning Corpus along with the Hugging Face Transformers library. Specifically, the BertForQuestionAnswering model, a task-specific implementation of BERT designed to predict answer spans within a given context, was employed (Wolf et al., 2020). This model is based on the bert-base-uncased pre-trained architecture, which consists of 12 transformer layers, 768 hidden units, and 12 attention heads. The 'uncased' configuration normalizes text by treating uppercase and lowercase letters identically (Devlin et al., 2019).

Hyperparameter optimization played a crucial role in achieving optimal performance during fine-tuning. Key hyperparameters were fine-tuned based on performance metrics such as training and validation loss. Each hyperparameter was carefully adjusted to balance effective learning and generalization. The fine-tuning process involved iterative weight adjustments over six epochs, minimizing errors on training examples while monitoring training and validation losses. An epoch refers to one complete pass through the entire training dataset, allowing the model to learn from all available examples in the data. The selected hyperparameters are summarized in Table 1.

Table 1: Hyperparameter Settings and Explanations

This table outlines the key hyperparameters used during the fine-tuning process for BERT and T5 models. Each hyperparameter, its value, and its purpose in optimizing the training process are detailed, highlighting choices made to ensure effective learning and generalization.

Hyperparameter	Value	Explanation
Epochs (NE)	6 (BERT), 3 (T5)	During training, the model iterated through the entire dataset for a total of six epochs (BERT) and three epochs (T5). Monitoring the training and validation losses showed that six epochs achieved a good balance between effective learning and generalization, ensuring the model converged without excessive overfitting.
Batch Size (BS)	8	The batch size of 8 was chosen to balance memory efficiency and stable learning. Smaller batches allow for more frequent weight updates, promoting faster results and introducing noise that can improve generalization.
Learning Rate (LR)	2e-5 (Bert), 3e-4 (T5)	These step sizes were used to update model weights during training, with the smaller value ensuring stable fine-tuning for BERT and the larger value optimizing convergence for T5.
Optimizer (WD)	AdamW	The AdamW was used to update the model's weights. AdamW incorporates weight decay for regularization, improving generalization during fine-tuning.

Despite adapting BERT through fine-tuning and hyperparameter optimization, its output often consisted of single words or phrases that lacked coherence and did not align well with the input questions. This outcome prompted the exploration of alternative models, such as T5, which was expected to better address the challenges of answering domain-specific questions in a more structured manner.

3.2.2 Fine-Tuning T5 for QA

To assess the comparative effectiveness of different language models in our QA framework, we incorporated the T5 (Text-to-Text Transfer Transformer) model alongside BERT. T5 represents a significant advancement in natural language processing (NLP) by treating every NLP task as a text-to-text problem, where both input and output are represented as text strings. This flexible framework enables T5 to handle diverse tasks, including retrieval, summarization, and question answering (Feng et al., 2022). T5 encoder-decoder architecture enables it to excel in generative tasks such as producing coherent and contextually relevant answers (Najafi & Tavan, 2022).

Multiple T5 models exist, ranging from T5-small to T5-XXL. We chose the T5-base model because it strikes a balance between computational efficiency and performance, making it suitable for adaptation using the Fine-Tuning Corpus of academic articles while remaining feasible to run on a CPU. We employed the T5ForConditionalGeneration variant of T5 in our methodology, which is well-suited for transforming questions into detailed textual answers.

The fine-tuning of T5 followed a process similar to that used for BERT, as summarized in Table 1. Harnessing the Hugging Face Transformers library and tracked using Weights & Biases, T5 was trained over three epochs with a learning rate (3e-4) and batch size (8). Despite these adjustments, the fine-tuned T5-base model's outputs like BERT's often consisted of disjointed words or phrases from the input question, lacking meaningful structure. These limitations highlighted the need for a more advanced approach. To address this, RAG was introduced, combining retrieval methods to supply the models with relevant information and improving the quality and accuracy of their responses.

3.3 Implementing Retrieval-Augmented Generation

To address the limitations observed during the fine-tuning of BERT and T5 for question answering, the RAG framework was implemented. RAG integrates retrieved information from a structured corpus with generative text capabilities, to produce more detailed and accurate answers to specific questions. This can be achieved either by a single LLM performing both retrieval and generation (a standalone model) or by using one LLM for retrieval and a different LLM for generation. This distinction is critical to understanding the comparative evaluation of approaches discussed later.

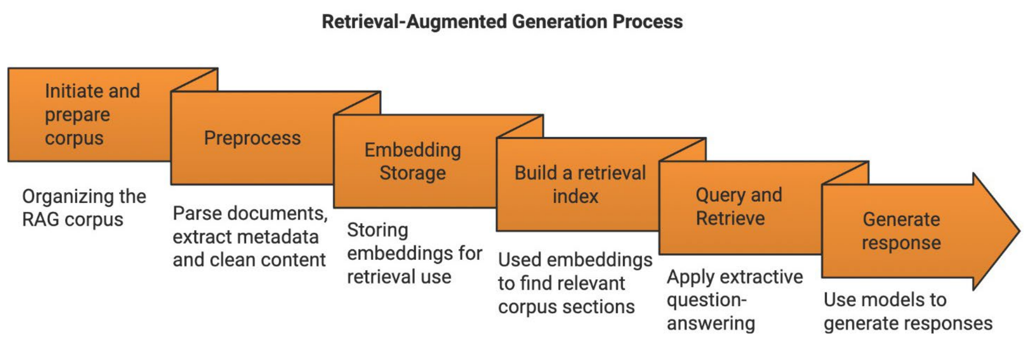

Figure 5: The Retrieval-Augmented Generation (RAG) Process
A high-level workflow for integrating retrieval with generation in large language models, highlighting steps such as corpus preparation, semantic embedding generation, and query-based augmentation for context-aware responses.

Figure 5 illustrates the key stages of this process, beginning with organizing and preprocessing the RAG corpus by parsing documents, extracting metadata, and cleaning content for downstream use. Semantic embeddings are generated and stored, enabling the construction of a retrieval index to efficiently identify relevant corpus sections during query processing. The retrieved content is subsequently incorporated into the generative pipeline, producing outputs informed by the retrieved context. The subsequent subsections detail each phase of this methodology, including embedding and structuring the corpus, implementing retrieval mechanisms, and integrating generative models into the RAG pipeline.

3.3.1 Embedding and Structuring the RAG Corpus

A critical pre-step in implementing RAG involved embedding the RAG Corpus into a structured and searchable format designed for retrieval and generation. Unlike the earlier XML processing step, which focused on organizing raw document structures, this process prepared the curated corpus specifically for embedding by transforming it into smaller, consistent, and retrievable units.

Embeddings are high-dimensional numerical representations of text, where each dimension encodes semantic features of the content. These representations allow models to compare and analyze text chunks based on their meaning rather than their surface-level wording (Wang & Dong 2020).

To generate embeddings, a Python script parsed the RAG corpus, extracting metadata such as document titles or IDs to maintain traceability. Sections and subsections were divided into smaller text chunks, each assigned a unique identifier (chunk_id) to ensure compatibility with the retrieval pipeline. These chunks preserved the hierarchical relationships and logical flow of the original PDF files, ensuring compatibility with the retrieval pipeline.

Once the corpus was structured, the text chunks were passed through the all-MiniLM-L6-v2 model from the Sentence-Transformers library to create embeddings. Each embedding captured the semantic features of its corresponding text chunk, enabling precise comparisons between queries and corpus content. Metadata with each embedding ensured retrieved results could be traced back to their source and contextualized during downstream processing.

The embeddings generated in this step were stored and prepared for retrieval using methods described in the next section. Figure 5 illustrates the key steps in structuring the RAG corpus for embedding and retrieval.

3.3.2 Retrieval Processes

Building on the embeddings created in the previous step, both the BERT and T5 pipelines employed semantic retrieval to locate the most relevant sections of the RAG corpus for domain-specific queries. This process corresponds to the Query and Retrieve phase in Figure 5, where embeddings are leveraged to identify relevant content for generation. These embeddings, generated using the Sentence-Transformers model (all-MiniLM-L6-v2), captured the semantic relationships between the query and text chunks, enabling accurate comparisons.

For the T5 pipeline, the embeddings were indexed in a FAISS (Facebook AI Similarity Search) system, a highly efficient tool for nearest-neighbor searches (Ljungquist et al., 2022). FAISS calculated the Euclidean distances between the query and corpus embeddings, where the Euclidean distance measures the straight-line distance between two points in the embedding space. These distances were converted into similarity scores using the formula:

$$Similarity\,Score \;=\; 1/(1\,+\,Distance)$$

Sections with similarity scores below 0.2 were excluded, and the top five most similar sections were retained. Each result was paired with metadata, including section titles, parent sections, document IDs, and chunk IDs, ensuring traceability and interpretability.

In contrast, the BERT pipeline did not use a pre-built index. Instead, cosine similarity metrics were dynamically computed during retrieval where higher cosine similarity values indicate greater alignment in semantic content (Wang & Dong, 2020). A threshold of 0.4 was applied to filter out less relevant passages, and the top five results were selected for further processing. Retrieved passages were paired with metadata, such as document titles and section names, ensuring traceability.

By employing these retrieval mechanisms, the RAG pipeline ensured that only the most relevant sections of the corpus were passed to the generative models for response generation.

3.3.3 Generation Processes

Once relevant passages were retrieved the BERT pipeline employed an extractive question-answering approach using the fine-tuned BertForQuestionAnswering model. The retrieved passages were concatenated into a single input context, which was tokenized together with the input query to prepare the data for the model. For each token in the input, the model calculated two scores: the likelihood that the token marked the start of the answer and the likelihood that it marked the end. These scores, called token-level logits, represent the model's confidence in the role of each token. By identifying the tokens with the highest start and end scores, the model predicted the most probable span of text to serve as the answer. This approach ensured that the output was directly grounded in the retrieved text.

In contrast, the T5 pipeline used a generative framework to produce answers from the retrieved passages. The retrieved chunks were formatted into a structured input, with each chunk labeled as a source and concatenated with the query. Instructions were included to guide the model in synthesizing a response based on the provided sources. The fine-tuned T5 model (T5ForConditionalGeneration) was then used to generate an answer, with decoding parameters carefully adjusted to enhance output quality. These parameters included beam search, which explores multiple possible answers to find the most likely one; repetition penalty, which discourages the model from repeating phrases excessively; and nucleus sampling, which selects words from the most probable options to ensure the response is coherent and focused. Despite T5's flexibility in generating natural language responses, its outputs often lacked coherence and readability.

The limitations of BERT's extractive approach and T5's generative model highlighted the need for a more advanced solution. To address this, GPT-4 and GPT-4o were integrated into the RAG pipeline. These models, part of OpenAI's Generative Pre-trained Transformer (GPT) series, represent significant advancements in natural language processing, utilizing transformer architectures designed for extended context handling and complex generative tasks (Rathje et al., 2023).

GPT-4, a multimodal model (capable of processing both text and images) provides superior performance to previous GPT models in areas such as complex reasoning and contextual understanding (Hirano et al., 2024). In contrast, GPT-4o (the "o" stands for optimized) focuses on efficiency, achieving faster inference speeds and lower computational demands through model compression and domain-specific fine-tuning. This optimization makes GPT-4o particularly suited for resource-constrained scenarios while maintaining high accuracy in specialized tasks (Samadi et al., 2024).

In this study, both GPT-4 and GPT-4o were used to evaluate their effectiveness in the RAG pipeline. By integrating these models, the generative phase aimed to overcome the limitations of BERT and T5, using GPT-4 and GPT-4o's advanced capabilities to produce responses that were expected to be more coherent, detailed, and aligned with domain-specific questions.

3.3.4 Prompt Engineering and Fine-Tuning Impact

Prompts played a significant role in framing how GPT-4 and GPT-4o interpreted the retrieved context and significantly influenced the quality and reliability of the generated responses. For the T5 pipeline, GPT-4 was employed with a highly structured and specific prompt that explicitly instructed GPT-4 to focus solely on the retrieved context generating detailed, logical, and comprehensive answers based exclusively on the provided information. It also included a fallback mechanism directing GPT-4 to state when the context was insufficient to answer the question.

In contrast, the initial prompt for the BERT pipeline, which used GPT-4o, was less stringent. While it instructed the model to base answers on the retrieved context, it lacked the explicit fallback mechanism and detailed formatting found in the T5-GPT-4 prompt.

To address these differences in prompt designs, both prompts were refined to ensure consistency and reduce variability in the quality of output between GPT-4 and GPT-4o. The revised prompts emphasized strict adherence to the retrieved information and introduced fallback instructions. For example, both prompts explicitly instructed the models to indicate when the provided context was insufficient to generate a meaningful answer. Additionally, parameters such as temperature (0.6) and a maximum token limit (5000) were consistently applied across both configurations, standardizing the conditions for generating responses.

Building on these refinements in prompt design, an additional experiment was conducted to evaluate the impact of fine-tuning on the retrieval phase. This experiment tested the standard (non-fine-tuned) T5-base model in the RAG pipeline, using GPT-4 for generation to isolate the effect of fine-tuning on retrieval quality. All configurations, including prompt design, decoding parameters, and evaluation metrics, were kept consistent to isolate the effect of fine-tuning during retrieval.

3.4 Evaluating the Models

To assess the quality of outputs generated by the LLMs in answering domain-specific questions about PEDs, two complementary evaluation methods were employed. First, expert evaluations were gathered through a structured questionnaire, focusing on subjective metrics such as accuracy, relevance, and scientific rigor. Second, cosine similarity analysis was conducted to objectively measure the semantic alignment between generated outputs and reference answers curated from a relevant corpus. This dual approach provided a comprehensive assessment, balancing subjective judgments with computational metrics.

3.4.1 Expert Evaluations

Expert evaluations were conducted using a structured questionnaire, a common and effective tool for collecting research data systematically. For this study, the questionnaire was designed to evaluate the quality of outputs generated by five LLMs and ChatGPT. The evaluation was divided into three surveys, each containing 10 text outputs, for a total of 30 outputs (see sample survey at Appendix A). These outputs were randomized to minimize potential bias. Each text output was evaluated on six metrics: accuracy, relevance, readability, scientific rigor, conciseness, and evaluator confidence. A Likert scale ranging from 1 (lowest) to 5 (highest) was used for the assessment. This systematic approach facilitates the comparison of responses and helps identify patterns and relationships within the data (Oates et al., 2022).

The questionnaire was created using Microsoft Forms and reviewed by the thesis supervisor to ensure clarity and alignment with the research objectives. It was distributed to six pre-selected domain experts employed by Dalarna University, comprising three faculty members, one post-doctoral researcher, and two PhD students. Participants were informed about the purpose of the study and the nature of their involvement prior to participation. Clear instructions on how the data would be used were provided both via email in advance and at the start of the survey. They were given one week to complete their evaluations, and responses were submitted anonymously through a secure link. Ethical standards were upheld by anonymizing all data and obtaining consent through an email invitation and a consent statement embedded in the survey.

To avoid bias, participants were instructed to evaluate each text independently without comparing outputs across models. Upon completion, six questionnaires were collected, providing structured feedback on all 30 text outputs. Quantitative analysis of the responses identified trends and patterns in model performance, offering insights into the strengths and weaknesses of the outputs. These findings guided improvements in fine-tuning methodologies for LLMs within the PED domain. Details of the analysis and its implications for the study are presented in subsequent sections of this thesis.

3.4.2 Cosine Similarity Analysis

The second method of evaluation involved measuring the cosine similarity between the generated outputs and reference answers manually curated from a relevant corpus. The analysis was performed using ChatGPT to process the texts and compute cosine similarity scores. Each generated output was paired with a reference answer from the corpus, and their similarity was evaluated to provide an objective assessment of how closely the generated responses align with established knowledge.

While expert evaluations captured human perceptions of quality, cosine similarity analysis provided a data-driven perspective. Together, these methods ensured a holistic evaluation of the model's outputs. The detailed results of both evaluations, including their implications for model fine-tuning and retrieval-augmented generation, will be presented in the results section.

4. Results

This section presents the outcomes of the methodology applied to construct and refine the system for generating accurate answers about PEDs using a RAG approach. The results are organized into distinct categories: the development of the corpus and preprocessing strategies, model fine tuning, retrieval and generation performance, and model output analysis. Each subsection highlights the key steps and their impact on the overall system, emphasizing the improvements made to ensure the system's reliability and relevance.

4.1 Descriptive Statistics and Keywords in Corpus

The curated corpus used for the RAG processes consists of 50 documents, including 49 peer-reviewed articles and one manually created dataset with 40 question-and-answer pairs. The corpus exclusively focuses on PEDs and includes publications from 2022 to 2024.

Manual reviews of the corpus addressed gaps caused by extraction errors in earlier iterations. This intervention increased the completeness and scope of the corpus, raising the number of sections by 74.8% from 726 to 1,269 (refer to Table 2). Similarly, the number of lines increased by 62.9%, from 4,799 to 7,817. These enhancements ensured the inclusion of critical content in the final corpus.

Table 2. Descriptive Statistics of the RAG Corpus
Summary of the corpus structure and preprocessing, including the number of documents, sections, and tokens before and after preprocessing, as well as details of chunking.

Number of XML/CSV files used in RAG corpus:	49/1
Number of lines of code including chunks	15161
Number of sections	1269
Number of tokens before preprocessing	388,491
Number of words: largest section (non-chunked)	2493
Number of chunks	4557
Number of words: average for chunked sections	50

Initial preprocessing highlighted challenges with tokenization, where domain-specific phrases such as "passive house" and "energy balance" were split into individual tokens, disrupting their semantic integrity. To resolve this, bigram and trigram tokenization strategies were implemented to preserve these multi-word phrases as single units. These adjustments ensured that key terminology remained intact and meaningful.

After preprocessing, the total token count dropped from 388,491 (see Table 2) in the original text to 227,135 in the processed text representing a 41.53% reduction. Similarly, the vocabulary size decreased from 28,111 to 12,508 unique tokens, reflecting a 55.5% reduction. The average length of a processed text section was approximately 179 words, although multiple sections exceeded BERT's 512-token limit, requiring chunking as outlined in the methodology section. The final RAG corpus consisted of 4,557 chunks, averaging approximately 50 tokens per chunk (see Table 2), with the largest chunk extending to 87 tokens.

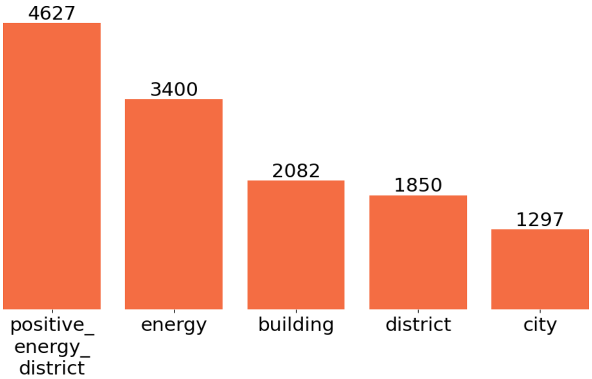

Figure 6. Top Five Keywords in the RAG Corpus

The five most frequent keywords in the corpus, with "positive_energy_district" dominating, reflecting the focus on PED-related content.

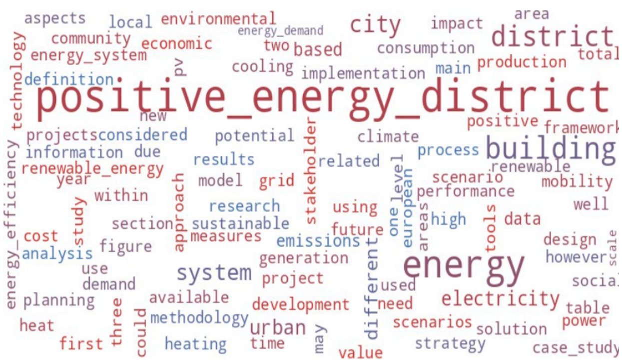

Figure 7. Word cloud of Common Words in RAG Corpus

The 100 most frequent words are shown, with larger font sizes indicating higher frequencies. Key terms include "energy", "district" and "positive_energy_district."

Key terms related to PEDs are prominent in the processed corpus, as shown in Figure 6, which lists the five most frequent keywords. The term "positive_energy_district" appears 4,627 times, reflecting the corpus's strong domain focus. Similarly, the word cloud in Figure 7 visualizes the 100 most frequent terms, with larger font sizes denoting higher frequencies. This representation further underscores the relevance of terms such as "energy" and "building" in the corpus. Together, these visuals highlight the corpus's suitability for answering PED-specific questions.

While preprocessing and refinement significantly improved the quality of the corpus, the impact on generation outputs depended on the model configuration, as explored in later sections.

4.2 Model Fine-Tuning Results

This subsection presents the results of fine-tuning and testing multiple models to evaluate their performance in answering PED-specific questions using RAG pipelines. Five pipelines were constructed, combining different retrieval and generation models, with each pipeline using either BERT or T5 for retrieval and GPT variants for generation. These pipelines were evaluated to assess their capacity to handle domain-specific content effectively.

The pipelines tested included:

1. **BERT**, fine-tuned on the corpus for both retrieval and generation.
2. **BERT-GPT4o**, using fine-tuned BERT for retrieval and GPT4o for generation.
3. **T5**, fine-tuned on the corpus for both retrieval and generation.
4. **T5-GPT4**, using fine-tuned T5 for retrieval and GPT4 for generation.
5. **NFT T5GPT**, which employed non-fine-tuned T5 for retrieval and GPT4 for generation.

4.2.1 BERT Model Fine-Tuning Results

The bert-base-uncased model was fine-tuned over six epochs using the BertForQuestionAnswering variant. During this process, training loss steadily decreased from 5.9333 in Epoch 1 to 2.0879 in Epoch 6, while validation loss declined from 5.1446 to 3.4617 as shown in Table 3. These reductions indicate that the model was progressively adapting to the fine-tuning corpus.

Table 3: Training and validation loss for the BERT model across six epochs.
The table demonstrates a steady reduction in training and validation loss.

Epoch	Training loss	Validation loss
1	5.9333	5.1446
2	4.7536	4.1862
3	3.9891	3.6278
4	3.3456	3.3739
5	2.7280	3.4490
6	2.0879	3.4616

The model's performance was evaluated as seen in Table 4 using precision, recall, and F1-score metrics for both the start and end tokens of answer spans. Precision measures the accuracy of predicted tokens as part of an answer, while recall assesses the proportion of actual answer tokens successfully identified by the model. The F1-score balances precision and recall to provide a single performance metric.

Table 4. Precision, Recall, and F1-Scores by Epoch
Performance metrics for the BERT model, highlighting improvements in identifying start tokens but limited gains for end token predictions.

Epoch	Precision Start	Precision End	Recall Start	Recall End	F1-score
1	0.0833	0.0357	0.0714	0.0714	0.0623
2	0.1750	0.0286	0.2500	0.0714	0.1233
3	0.1750	0.0286	0.2500	0.0714	0.1233
4	0.1750	0.0286	0.2500	0.0714	0.1233
5	0.1750	0.0308	0.2500	0.0769	0.1249
6	0.1555	0.0308	0.2000	0.0769	0.1095

The training results highlight that the BERT model performed better at identifying the start tokens of answers compared to end tokens. Precision for start tokens improved consistently from 0.083 in Epoch 1 to a peak of 0.175 by Epoch 5, whereas precision for end tokens remained low, fluctuating between 0.035 and 0.031 across epochs. Recall for start tokens also improved, increasing from 0.071 in Epoch 1 to 0.250 by Epochs 2–5, but recall for end tokens showed minimal improvement, ranging from 0.071 to 0.077. Consequently, the overall F1-score remained modest, starting at 0.062 in Epoch 1 and peaking at 0.124 in Epoch 5, before slightly declining to 0.109 in Epoch 6.

4.2.2 T5 Model Fine-Tuning Results

The T5 model demonstrated a different learning pattern during fine-tuning. Training loss showed consistent improvement as detailed in Table 5, starting at 0.0454 during steps 50–100, decreasing to 0.0264–0.0213 by steps 150–200, and stabilizing at 0.0217 by step 250. Table 6 shows that the final global training loss was recorded at 0.0304, with a total training runtime of 373.23 seconds. During this time, the model processed an average of 5.836 samples per second, with each optimization step taking approximately 0.731 seconds.

Table 5. T5 Training Loss

Training loss across optimization steps, showing a steady decrease and stabilization in later steps, indicating effective adaptation to the corpus.

Step range	Training loss
50 – 100	0.0454 – 0.0370
150 - 200	0.0264
250	0.0217

Table 6. T5 Fine-Tuning Performance Metrics

Metrics include training runtime, samples processed per second, and final global training loss.

Metrics	Value
Final global training loss	0.0304
Total training runtime (seconds)	373.23
Samples processed per second	5.836
Optimization Steps per second	0.731

4.3 Evaluation of the RAG processes

The fine-tuning results for both BERT and T5 highlight the need for effective retrieval and generation mechanisms to support accurate and contextually appropriate answers. To address these challenges the RAG pipelines were evaluated, beginning with the embedding process, which forms the foundation for effective information retrieval.

4.3.1 Results from the Embedding Process

The embeddings for the RAG corpus, used in both the BERT and T5 retrieval pipelines, were generated using the SentenceTransformer model (*all-MiniLM-L6-v2*). For the T5 retrieval pipeline, embeddings were indexed with FAISS, enabling efficient nearest-neighbor searches. This process successfully incorporated all 4,557 embeddings, verifying the completeness of the embedding process. In contrast, the BERT pipeline computed cosine similarity directly between the query embedding and corpus embeddings without using FAISS indexing. To validate the retrieval process, a test query confirmed that retrieval operated as expected, with the closest embedding having a distance of 0.0 (self-similarity) and subsequent neighbors exhibiting small distances (e.g., 0.35–0.37), indicating meaningful clustering of similar content.

To assess the quality and distribution of the embeddings themselves, cosine similarity was computed for a small subset of the RAG corpus after embedding. The results revealed a mix of closely and loosely related chunks, with cosine similarity scores ranging from 0.75 (high similarity) to 0.15 (low similarity). This variability demonstrates that the model effectively distinguishes between semantically similar and dissimilar text, suggesting that the embeddings are not overly redundant.

4.3.2 Results of Refining the Retrieval Processes:

Building on the embedding process, the retrieval phase was analyzed to determine how effectively the pipelines returned relevant sections from the corpus. An intriguing observation emerged during testing in which both BERT and T5-based retrieval systems consistently returned the same top-K sections in identical order for a range of test queries, including synthetic ones. This consistency occurred regardless of whether the T5 model was fine-tuned. As previously noted, both pipelines relied on the same SentenceTransformer model to generate corpus embeddings, which explains the uniformity in retrieved sections. However, while the retrieved sections were identical, slight differences in similarity scores arose due to the computation methods in which BERT used cosine similarity directly and T5 employed FAISS distances converted into similarity scores. To further evaluate retrieval performance, tests included both domain-relevant and out-of-domain queries.

For example, an illustrative query—'What is the habitat of Polar Bears?'—was used to test the retrieval process. Despite being unrelated to the PED-focused corpus, the query returned sections with relatively high similarity scores (0.40–0.42). While the analysis section explores a possible reason for this outcome, it highlighted a limitation of embedding-based similarity measures when handling out-of-domain queries. To address this limitation, synthetic PED-specific questions were introduced as a controlled evaluation method. These questions were carefully designed to align with the corpus domain, enabling a focused assessment of the retrieval system's ability to identify semantically meaningful sections relevant to PEDs. By grounding all test queries within the scope of the corpus, this approach ensured a more realistic and reliable evaluation of retrieval performance.

4.3.3 Results from Prompt Development in the Generation Process

While retrieval improvements enhanced the relevance of retrieved sections, they alone did not guarantee grounded answers in the generation phase. Early testing revealed that GPT-4 and GPT-4o models often generated plausible but unsupported responses, as seen in the out-of-domain polar bear question, where retrieved sections were unrelated to habitats and provided no basis for an accurate answer. To address this, we refined the generation prompt in stages to explicitly guide the model to rely solely on the retrieved content.

Table 7 illustrates how these refinements progressed, with the first two iterations of the prompt generating nearly identical outputs that were not based on the retrieved text. However, the final iteration successfully directed the model to focus solely on the retrieved context, ensuring responses were detailed, specific, and comprehensive while transparently addressing gaps in the provided information and avoiding unsupported generalizations.

Table 7. Iterative Development of GPT Prompts

Refinements in the GPT generation pipeline prompts, demonstrating improvements in responses to the question "What is the habitat of Polar Bears?" with excerpts from the generated outputs.

Iteration	Prompt	Representative Output
1st	*"Based on the following context, answer the question:"*	*"…They are well-adapted to survive in the cold environment of the Arctic."*
2nd	*"Answer the following question based on the information provided:"*	*"…They are well-adapted to survive in the cold environment of the Arctic."*
3rd	*"Please provide a detailed and comprehensive answer to the following question based on the information provided:"*	*"…The habitat of polar bears is closely tied to sea ice as they rely on it for hunting seals…."*
4th (Final)	*"Answer the following question based on the context provided. Focus solely on the retrieved information and ensure the response is detailed, specific, and comprehensive. If the retrieved texts lack sufficient detail, provide a specific and logical response based on the available context without making unsupported generalizations."*	*"The provided context does not contain any information about the habitat of Polar Bears…."*

The final refined prompts also influenced how models using GPT-4 or GPT-4o framed their responses to authentic questions about PEDs, particularly when the retrieved context lacked sufficient detail. For example, when asked, *"What were the results of retrofitting buildings to comply with the BR10 standard in terms of energy demand reduction in Odense, Denmark?"* the TF-GPT4 model responded, *"The documents provided do not specify the results of retrofitting buildings to comply with the BR10 standard in terms of energy demand reduction specifically in Odense, Denmark. However... the text suggests that retrofitting buildings to correspond with the latest Danish regulation standard, such as BR10, results in a reduction of* *space and total heat demand. The exact amount... is not specified in the provided context, except that it would be... around many megawatt-hours (MWh) per year..."* explicitly acknowledging the limitations of the retrieved content. Similarly, for the question, *"What would be the key technologies for PED development in 5 years?"* BERT-GPT4o prefaced its answer with, *"Based on the provided documents, several key technologies are likely to be crucial for the development of Positive Energy Districts (PEDs) in the next five years. These include..."* signaling a reliance on available content without unsupported speculation.

Interestingly, ChatGPT, even without such prompts, received higher confidence ratings in the expert survey for answers that did not explicitly reference the provided context. These observations provide a foundation for the expert evaluations discussed in the next section, which examine the perceived effectiveness and reliability of the generated responses.

4.4. Results of Expert Evaluation

To evaluate the performance of different language models, we conducted a survey with six domain experts in PEDs. The experts assessed the five pipelines described in Section 4.2, along with ChatGPT4o, based on six key criteria: relevance, readability, logic, conciseness, confidence and accuracy.

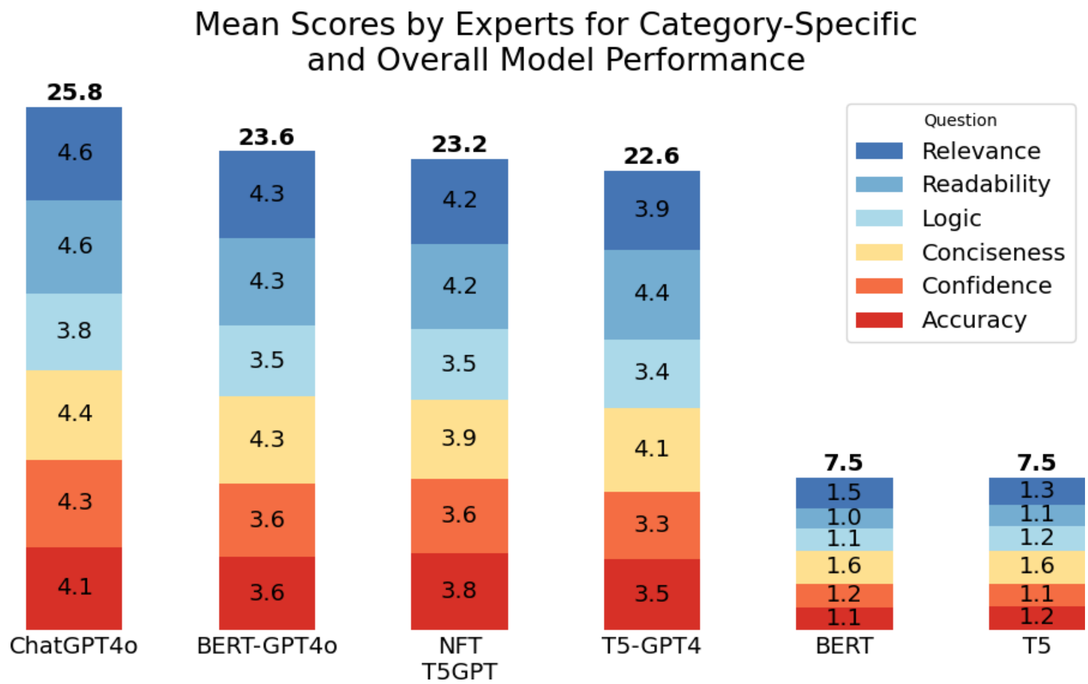

Figure 8. Mean Scores by Experts for Model Performance
Average scores across six evaluation criteria, with ChatGPT4o achieving the highest overall score (25.8), followed by BERT-GPT4o (23.6) and NFT T5GPT (23.2). Standalone BERT and T5 models scored significantly lower (7.5 each).

According to the results of expert evaluations seen in Figure 8, ChatGPT4o achieved the highest overall score of 25.8, demonstrating strong performance across all evaluated categories. Not only did it achieve the highest overall score, but it also outperformed all other models in each individual metric. Its best scores were in relevance (4.6) and readability (4.6), followed closely by conciseness (4.4), confidence (4.3), and accuracy (4.1). Although it scored slightly lower in logic (3.8), it still outperformed the other models in this category.

BERT-GPT4o scored 23.6, with notable strengths in relevance, readability, and conciseness (4.3 each), though it underperformed in logic (3.5) and confidence (3.6). NFT T5GPT followed closely at 23.2, with balanced scores across categories, performing best in relevance and readability (4.2 each).

T5-GPT4 scored 22.6, performing strongest in readability (4.4) but struggling in logic (3.4) and confidence (3.3). Non-GPT models, including BERT and T5, scored significantly lower, each achieving an overall score of 7.5. These pipelines particularly struggled in accuracy (BERT: 1.1, T5: 1.3) and logic (BERT: 1.0, T5: 1.2), demonstrating substantial limitations in producing precise outputs.

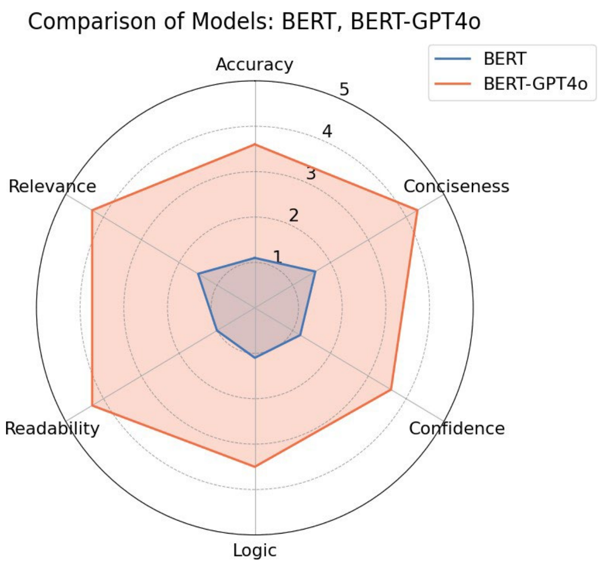

Figure 9. Radar Plot Comparing BERT and BERT-GPT4o
Comparison of evaluation metrics, showing BERT-GPT4o (orange) outperforming BERT (blue) across all categories.

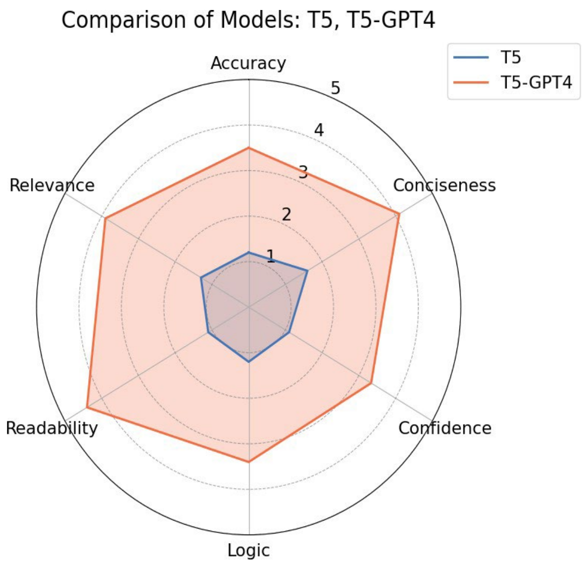

Figure 10. Radar Plot Comparing T5 and T5-GPT4
Evaluation metrics for T5-GPT4 (orange) indicate superior performance over T5 (blue), though challenges remain in accuracy, logic, and confidence.

The radar plots in Figures 9 and 10 highlight the significant improvements achieved by integrating GPT into the pipelines. BERT-GPT4o and T5-GPT4 demonstrated average performance increases of 330% and 300% in readability, respectively, over their non-GPT counterparts. Similarly, accuracy scores improved by 227% for BERT-GPT4o and 192% for T5-GPT4. Other metrics, including relevance and conciseness, also saw substantial enhancements. These results underline the transformative role of GPT models in achieving balanced and higher-quality outputs across all evaluation categories

4.5 Objective Analysis of Model Outputs Using Cosine Similarity

Cosine similarity was used to evaluate the semantic alignment between model-generated responses and reference answers. Scores closer to 1 indicate stronger alignment, while values closer to 0 suggest weaker alignment. This metric provides a data-driven complement to expert reviews, offering an objective measure of semantic accuracy across six language models: BERT, BERT-GPT4o, T5, T5-GPT4, NFT T5GPT, and ChatGPT.

4.5.1 Cosine Similarity with Reference Answers

Figure 11 visualizes the results, highlighting distinct patterns in the models' ability to produce semantically similar responses to reference answers for three PED-related questions.

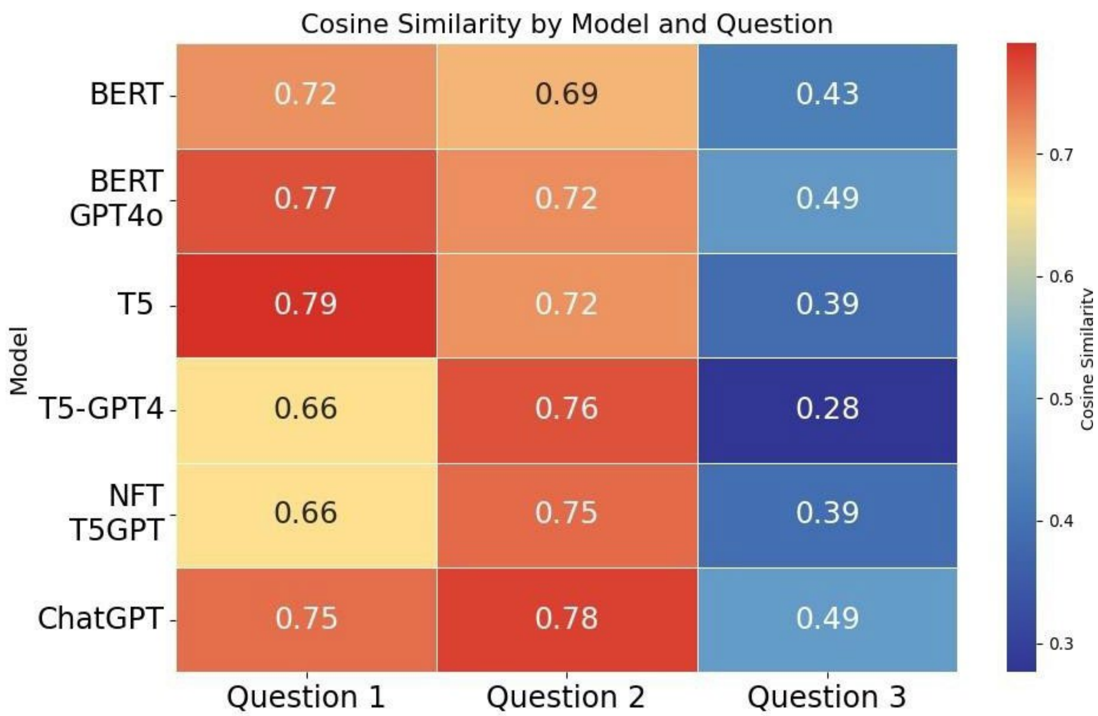

Figure 11. Cosine Similarity by Model and Question
This heatmap shows cosine similarity scores between model outputs and reference answers for three questions. Higher scores (red) indicate greater alignment, while lower scores (blue) reflect less similarity.

For the first question, *“What lessons can be learned from Norway's implementation of nine PED ventures that could be applied to other European cities considering similar projects?”,* cosine similarity scores ranged from 0.66 (T5-GPT4 and NFT T5GPT) to 0.79 (T5), with BERT-GPT4o (0.77) and ChatGPT (0.75) performing comparably well. These scores highlight that most models achieved moderate to high semantic alignment on this question.

The second question, *“Why is Vallastaden district in Linköping, Sweden suitable for being a PED?”* resulted in the most consistent performance among all models, with cosine similarity scores ranging from 0.69 (BERT) to 0.78 (ChatGPT). The relatively high scores and the narrow 0.09 range between the highest and lowest values suggest that the models performed similarly on this question and produced responses closely aligned with the reference answer.

The models outputs in the third question, *“How does the Stockholm Royal Seaport plan to achieve its goal of becoming fossil-free and climate-positive by 2030, and what challenges might they face in this endeavor?”*, showed more significant variation, with scores ranging from 0.28 to 0.49. Both ChatGPT and BERT-GPT4o tied for the highest score at 0.49, while T5-GPT4 had a much lower score of 0.28. This significant drop in performance suggests that this question posed a greater challenge across all the six models.

Following the analysis of the individual question results, the bar graph in Figure 12 provides a summary of the model's general performance across all three questions. It illustrates the average cosine similarity scores, allowing for a clear comparison of how each model performed.

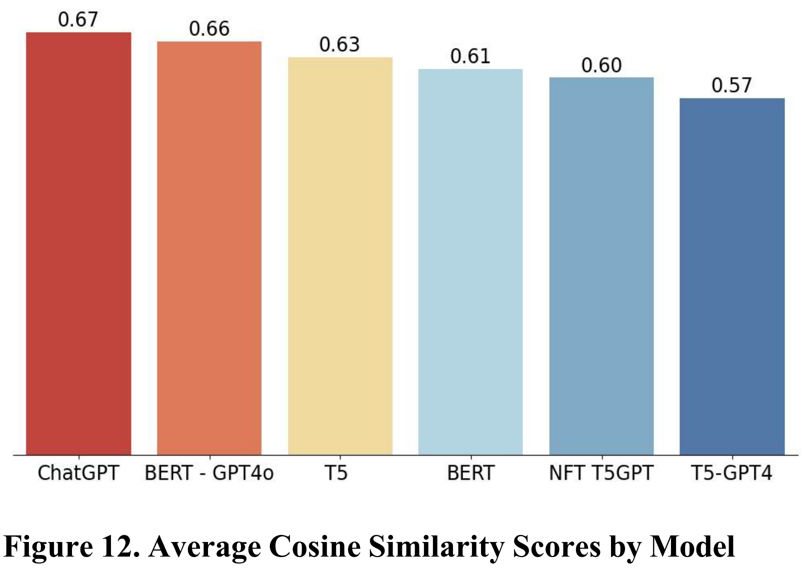

Figure 12. Average Cosine Similarity Scores by Model
This bar chart shows the average cosine similarity scores for six models, indicating their alignment with reference answers across domain-specific questions.

ChatGPT achieved the highest average score of 0.67, reflecting its consistent performance across all questions. BERT-GPT4o follows closely with an average score of 0.66, demonstrating strong alignment, although with slight variation compared to ChatGPT. Models like T5 and BERT scored moderately, with averages of 0.63 and 0.61, respectively, falling short of the top two models. NFT T5GPT recorded a slightly higher average score of 0.60 than T5-GPT4. T5-GPT4, with the lowest average score of 0.57, struggled the most in producing responses that closely matched the reference answers.

4.5.2 Cosine Similarity for Synthetic Questions

To assess the models' ability to handle out-of-context questions, we introduced a set of synthetic prompts. These questions explored topics or scenarios absent from the corpus, requiring the models to generate answers without access to relevant retrieved context. The cosine similarity scores for these questions reflect how closely the models' outputs aligned with each other, highlighting patterns in generative behavior rather than accuracy against a reference.

Table 8. Comparison of Model Scores for Sample Synthetic PED Question
Scores assigned to different model combinations for the synthetic PED-related question:
"What innovative energy solutions are being applied in the Dalarna neighborhood in Sweden to ensure it becomes a plus energy neighborhood?"

Models being compared	Score
BERT and T5	0.744
T5-GPT4 and NFT T5GPT4	0.806
BERT-GPT4o and T5-GPT4	0.5402
BERT-GPT4o and ChatGPT	0.2027

When comparing the performance of GPT-enhanced pipelines (e.g., BERT-GPT4o and T5-GPT4) to their non-GPT counterparts (e.g., BERT and T5), results reveal key differences in alignment and response behavior. As shown in Table 8, the T5-GPT4 and NFT T5-GPT4 pipelines achieved the highest cosine similarity score of **0.806** for the synthetic question: "What innovative energy solutions are being applied in the Dalarna neighborhood in Sweden to ensure it becomes a plus energy neighborhood?" This indicates an expected strong alignment between these GPT-augmented models.

The output from BERT and T5 also demonstrated a relatively high cosine similarity score of **0.744**, indicating strong alignment for the synthetic question, even without GPT enhancements. In contrast, the BERT-GPT4o and T5-GPT4 combination yielded a lower similarity score of **0.5402**, indicating less alignment when generating responses to synthetic queries.

Interestingly, when examining the models that performed best in expert evaluations (BERT-GPT4o and ChatGPT), their cosine similarity score for the synthetic Dalarna neighborhood question was just **0.2027**. Both models leverage GPT-4o, yet their divergent generative behavior reflects differences in how they prioritize responses. For example, the RAG-enhanced BERT-GPT4o pipeline generated outputs explicitly guided by the prompt outlined in Table 7, resulting in a concise response that acknowledged the absence of sufficient contextual information. In contrast, ChatGPT produced a more detailed response, which included plausible but unverifiable content.

5. Analysis of Results

This study evaluated the effectiveness of retrieval-augmented generation (RAG) pipelines across multiple dimensions, highlighting the interplay between corpus construction, model fine-tuning, prompt refinement, and embedding consistency. The discussion explores the key findings, including the success of GPT-enhanced pipelines in addressing domain-specific and synthetic questions, the limitations of non-GPT models, and the role of structured corpora in ensuring retrieval quality. Broader implications for Positive Energy Districts (PEDs) and other specialized domains are also considered.

5.1 Corpus Construction and Preprocessing

The corpus construction played a critical role in ensuring relevance and retrieval effectiveness. Chunking preserved the logical flow of content while adhering to the token limitations of the model. This segmentation strategy ensured the system could effectively retrieve precise and relevant information during question-answering tasks. The structured corpus, enriched with metadata, provided contextual anchors for embeddings and improved traceability for downstream tasks. This structure allowed the retrieval process to display specific articles and sections, validating the quality of the retrieval and supporting more grounded answer generation. While standard preprocessing steps such as stop word removal, normalization, tokenization, and lowercasing were employed, no clear benefits or drawbacks were observed in the context of retrieval or generation. This suggests that preprocessing may have limited relevance when using advanced GPT models like GPT-4 or higher in RAG systems. Such models are designed to process text contextually and can retain semantic understanding even when working with raw, unprocessed input (Kung et al.,2023).

5.2 Model Fine Tuning and Model-Specific Observations

The fine-tuning results for BERT and T5 on the corpus revealed distinct learning behaviors and performance outcomes. BERT demonstrated consistent improvement in both training and validation losses showcasing its ability to adapt to the corpus during fine-tuning. However, it faced difficulty in accurately identifying the start and end tokens for the answer spans. This ultimately impacted its ability to generalize effectively as reflected in the F1 scores. In contrast, T5 had a clear and efficient learning pattern during fine-tuning. While it started with a relatively high loss, the loss steadily decreased in the later stages of the finetuning, suggesting that the model was able to quickly adapt to the corpus and learn from it. The process was efficient with the model processing a good number of samples per second allowing it to complete the fine tuning in a reasonable amount of time. While both models improved during fine-tuning, they struggled with generating accurate answers. BERT showed struggles with precision and recall while having specific issues with identifying end tokens. T5, despite being more efficient in training, faced similar challenges. The results suggest that both models need to be enhanced for better performance in domain-specific tasks.

Fine-tuning was expected to improve the models' performance. However, the non-fine-tuned T5-GPT model performed almost identically to its fine-tuned counterpart, as demonstrated by similar expert evaluation results and cosine similarity scores. This underscores the generative power of GPT models combined with a well-structured corpus, suggesting that fine-tuning may be less critical for such setups. These findings align with the broader observation that high-quality generation models, paired with effective retrieval mechanisms, can often outweigh the incremental gains from fine-tuning.

5.3 Prompt Refinement

Prompt refinement significantly improved output quality for out-of-domain questions, directing the model to rely on retrieved sections and reducing hallucinations. The impact of prompt refinement on synthetic domain-specific questions could not be explicitly measured, as the prompt had already been finalized before synthetic questions were introduced. While this limited our ability to compare iterations, the refined prompt's directive to focus on retrieved content and avoid unsupported generalizations appeared to enhance answer grounding for both synthetic and domain-specific questions.

5.4 Embedding Consistency in Retrieval Processes

Embedding consistency emerged as a crucial factor in ensuring retrieval accuracy. The embedding consistency observed in retrieval, where models retrieved the same sections in identical K-order, underscores that the choice of retrieval model (e.g., BERT vs. T5) is less impactful than the quality of corpus embeddings. This finding reinforces the importance of creating high-quality embeddings as a foundation for retrieval-augmented generation (RAG) systems, particularly when dealing with large, structured corpora.

5.5 Discussion of the Expert Evaluation

The expert evaluation revealed significant differences in the performance of various LLMs across key criteria: accuracy, relevance, readability, logic, conciseness, and confidence. ChatGPT4o demonstrated the highest overall performance, excelling in accuracy, relevance, conciseness, and readability, with solid scores in logic and confidence. Although its readability score was slightly lower than expected, it still surpassed other models, showcasing its reliability for tasks requiring high-quality and accurate outputs.

BERT-GPT4o performed well in relevance, logic, and accuracy but scored lower in readability and conciseness. This indicates strong content generation capabilities but suggests room for improvement in producing more concise and accessible outputs. Its balanced performance makes it a strong alternative, especially when accuracy is a priority. Similarly, NFT T5GPT showed relatively even scores across all categories, without excelling in specific areas. This moderate but consistent performance suggests its suitability for general-purpose tasks requiring adaptability rather than specialization.

T5-GPT4 stood out for its strong logic score but scored lower in readability and confidence, limiting its effectiveness for producing clear and dependable outputs. This model may be best suited for tasks emphasizing logical reasoning over interpretability. The comparison of models combining retrieval and generation (BERT-GPT4o and T5-GPT4) with those relying solely on retrieval or generation components (BERT and T5) further highlighted the advantages of advanced generative models. BERT and T5 showed significant limitations when used independently, reinforcing the importance of integrating generative models like GPT to enhance overall performance.

5.6 Evaluation of Cosine Similarity

We evaluated the semantic alignment between model-generated responses and reference answers using cosine similarity. Models were also tested on their ability to handle out-of-context questions, revealing notable differences in performance across three PED-related questions.

For question 1, T5 demonstrated the strongest semantic alignment, while T5-GPT4 and NFT T5GPT struggled the most, reflecting difficulties in processing the question. In contrast, the outputs to question 2 showed more consistent performance, with ChatGPT emerging as the top performer, followed closely by T5-GPT4. The relatively narrow score range (0.69–0.78) suggested a general alignment with reference answers.

The third question on the Stockholm Royal Seaport's fossil-free and climate-positive goals was more challenging for all models, with ChatGPT and BERT-GPT4o achieving the highest alignment despite lower overall scores.

Across all questions, ChatGPT demonstrated the most consistent performance, maintaining the highest average cosine similarity score. BERT-GPT4o followed closely, with T5 and BERT scoring moderately. NFT T5GPT and T5-GPT4 performed lower, indicating weaker semantic alignment. Despite these results, T5 and BERT outputs were often rated poorly by experts for readability and interpretability. This disconnect highlights that high semantic similarity does not guarantee clear or actionable outputs.

For synthetic questions, cosine similarity revealed key differences in how pipelines aligned with one another. NFT T5GPT4 and T5-GPT4 as well as BERT and T5 showed relatively high semantic alignment in their responses to the synthetic question about PEDs in Dalarna. In contrast, BERT-GPT4o and ChatGPT4o had the least similar output, with BERT-GPT40 providing context-restricted responses, explicitly acknowledging when information was insufficient. However, ChatGPT provided a detailed and plausible yet unverifiable response. This transparency aligns well with retrieval-augmented pipelines designed for accuracy and interpretability, making them better suited for applications where traceability is critical.

5.7 ChatGPT Performance and Web Search Functionality

ChatGPT consistently delivered the most effective outputs, as reflected in expert evaluations and cosine similarity scores, demonstrating its ability to excel even without access to the structured RAG corpus. During the study, OpenAI introduced functionality enabling ChatGPT to search the web when it lacked sufficient context to answer a question. This advancement allowed ChatGPT to handle domain-specific questions more effectively, often retrieving and synthesizing information beyond its training data.

ChatGPT's performance on synthetic questions exceeded expectations but revealed limitations. While it often provided reasonable and semantically aligned answers, it sometimes produced hallucinated responses, synthesizing accurate details into unsupported narratives. For example, when asked about Positive Energy Neighborhoods (PENs) in Dalarna, ChatGPT combined accurate details from online sources into a misleading conclusion, suggesting the existence of PENs in the region without support from the corpus or evidence. This highlights a limitation: GPT models can combine factual elements into misleading conclusions without rigorous grounding.

Despite such limitations, ChatGPT's adaptability and continuous evolution highlight its strength as a dynamic generative tool. Its ability to generate high-quality responses while flexibly accessing external information positions it as a valuable complement to retrieval-based systems, combining strong language capabilities with enhanced functionality.

5.8 Broader Implications

In the context of PEDs, this study emphasizes the value of structured corpora and embedding-based retrieval for streamlining access to fragmented PED-related literature. Such systems can enable researchers and policymakers to extract actionable insights on energy-efficient practices, automate the synthesis of best practices, identify gaps in current implementations, and guide collaborative initiatives across cities. These applications are particularly relevant for scaling sustainable energy solutions and fostering international collaboration.

More broadly, the findings highlight the foundational role of a well-structured corpus and high-quality embeddings in RAG systems. These elements underpin the effectiveness of RAG pipelines, while advanced generative models like GPT add significant value by enhancing the quality and coherence of generated responses. However, this study also underscores the notable strengths of standalone generative models like ChatGPT, which consistently outperformed RAG pipelines in expert evaluations and cosine similarity. ChatGPT's ability to generate accurate and coherent responses without relying on a structured corpus illustrates its potential as a tool for knowledge synthesis, particularly in domains where dynamic adaptability and broad generalization are key.

The principles demonstrated in this study extend beyond PEDs to other specialized fields such as healthcare, legal research, or environmental planning. In these domains, RAG systems can integrate domain-specific information to produce reliable and explainable outputs. Conversely, the dynamic and continuously improving capabilities of generative models like ChatGPT can complement or even surpass RAG systems in scenarios that demand generalizable reasoning or rapid adaptability. Balancing the strengths of both approaches offers a promising direction for advancing intelligent information systems in specialized and cross-disciplinary contexts.

6. Conclusion

PEDs represent an innovative approach to addressing urban energy challenges, aiming to create sustainable, energy-positive environments that reduce greenhouse gas emissions. As PED implementation expands across Europe, effective tools for information exchange and decision-making are crucial to support multi-stakeholder engagement and ensure the successful realization of these projects. This thesis has explored the potential of QA frameworks, particularly those utilizing RAG, in addressing the complexities of PED implementation.

The research identified a gap in QA frameworks specifically tailored to PEDs, presenting an opportunity to develop a specialized, chat-based platform for stakeholders. Results demonstrate the effectiveness of RAG in enabling large language models to address domain-specific queries related to PEDs. GPT-enhanced pipelines consistently outperformed standalone models in expert evaluations and semantic alignment, generating more accurate and relevant outputs. However, standalone models like BERT and T5, despite high cosine similarity scores, struggled with readability and coherence, revealing a disconnect between semantic alignment and practical usability. ChatGPT, while versatile and effective without a structured corpus, occasionally produced unsupported narratives, highlighting the trade-offs of generative richness versus grounding.

These findings underscore the importance of structured corpora, rigorous evaluation methods, and a balance between retrieval-based and generative approaches. Further research could explore adaptive prompts, hybrid pipelines, and refined evaluation metrics to enhance user-centered outputs.

In conclusion, this thesis demonstrates that advanced QA frameworks, particularly those based on RAG and generative models, can play a pivotal role in advancing the development and implementation of PEDs. By improving information retrieval and decision-making processes, such frameworks can significantly contribute to urban sustainability efforts in Europe, fostering collaboration among diverse stakeholders and supporting the creation of energy-positive urban environments.

6.1 Limitations

Several limitations should be acknowledged. First, the number of expert reviews was limited to six, which amplified the influence of individual reviewers. Notably, results shifted when the number of reviewers increased from five to six, demonstrating the sensitivity of rankings to sample size.

Second, named entity recognition (NER) was not conducted during corpus preprocessing. Incorporating NER could have enhanced the effectiveness of fine-tuning by explicitly tagging critical entities, potentially improving retrieval and generation accuracy. Finally, while prompt refinement demonstrated clear benefits, we did not systematically measure its impact across all question types, limiting our ability to generalize its effectiveness.

Although the corpus was centered on PEDs and consisted of 49 peer-reviewed articles and 40 question-and-answer pairs, it may not fully represent the complete range of PED-related content. Its reliance on publications from 2022 to 2024 further limits coverage of relevant literature, potentially restricting the model's ability to answer questions requiring historical context or deeper insights into the evolution of PEDs.

6.2 Future Research

This study highlights several areas where RAG systems and domain-specific applications can be optimized and expanded. Future research could focus on developing tools and platforms designed specifically to address the challenges and requirements of PEDs. A chat-based platform for PED stakeholders could facilitate better interaction and collaboration, offering real-time communication tools, personalized assistance, and support for informed, data-driven decision-making.

In addition, tailored datasets and specialized knowledge bases could be developed to capture PED-specific terminology, regulatory requirements, and real-world challenges. These knowledge bases could include comprehensive data on environmental, technical, regulatory, and societal aspects, enabling more accurate and nuanced answers to PED-related inquiries. Future studies should explore how to effectively integrate such resources into existing QA systems, potentially through specialized knowledge graphs or databases. This integration could enhance domain expertise and provide researchers and policymakers with actionable insights on energy-efficient urban planning and implementation.

The expert evaluations in this study were limited by a small reviewer pool, which may have amplified the influence of individual reviewers. Expanding the sample size in future studies could help ensure a more representative evaluation of model performance. Additionally, future research could explore new evaluation metrics to better evaluate model effectiveness in addressing practical, real-world tasks. For instance, metrics like explainability—assessing how well a model's reasoning can be understood by users—could be particularly valuable in contexts requiring transparency, such as policy or regulatory compliance. Domain-specific metrics tailored to fields like energy policy or urban planning could further refine the evaluation process, providing a deeper understanding of the models' effectiveness in real-world applications.

A preliminary experiment conducted during this study suggested that preprocessing steps such as stop word removal and normalization may not significantly impact retrieval or generative accuracy for advanced GPT models. Initial findings indicated that embeddings derived from raw text performed comparably to those from preprocessed text, as long as token limitations were managed effectively. Future research could formally investigate these observations by comparing the retrieval quality and generative accuracy of embeddings generated from raw versus preprocessed text. This would clarify whether traditional preprocessing pipelines remain relevant when working with advanced LLMs in RAG systems.

Generative models like GPT demonstrated strong adaptability in this study, but further research should examine the impact of prompt adjustments on performance, transparency, and user trust. Instructing models to cite specific sources from retrieved documents could enhance grounding and credibility by enabling users to confirm the accuracy of the information provided, which is particularly important in fields like healthcare or regulatory compliance. Hybrid approaches that combine structured corpora with real-time access to external information (e.g., web searches) could also be explored. These approaches could balance retrieval accuracy with the detailed and nuanced answers informed by contextual information that generative models can provide. Additionally, research could investigate prompts designed to adapt outputs to user needs, balancing conciseness, comprehensiveness, and technical depth for diverse applications.

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Appendix A: Expert Survey Questions (Survey 1 of 3)

Evaluating the Quality of Texts Related to Positive Energy Districts (PEDs) - General 1

This form is used to collect feedback about texts relating to Positive Energy Districts

* Required

Information about the survey

Researchers: Michael Oppenheimer, Vivian Nanyka
Institution: Dalarna University
Contact Information: mom@du.se, Advisor: mea@du.se (Mengjie Han)

Purpose of the Study:
The purpose of this study is to evaluate the quality of responses generated by various tools in addressing questions about Positive Energy Districts (PEDs). Your feedback will help improve tools and methodologies in the field.

What Participation Involves:
You will review 10 texts related to PEDs and provide ratings on metrics such as accuracy, readability, and scientific quality.
The process should take approximately 30 minutes.

Voluntary Participation:
Participation is voluntary, and you may withdraw at any time without providing a reason or facing any consequences.

Confidentiality:
Your responses will be anonymized and securely stored. No identifying information will appear in any publications or presentations resulting from this research.

Potential Risks and Benefits:
There are no anticipated risks. By participating, you will contribute to advancing research in PED-related applications.

Consent:
By proceeding, you confirm that you:

1. Understand the purpose of this study and what participation involves.
2. Agree to provide feedback on the provided texts.
3. Understand you can withdraw at any time without consequences.

If you have questions, please don't hesitate to contact me at mom@du.se.

1. Do you agree to participate in this evaluation for a master's thesis project? Your responses will be anonymous, confidential, and used solely for research purposes. Participation is voluntary, and you may withdraw at any time without penalty.
*

- Yes, I agree
- No, I do not agree

Next

Evaluating the Quality of Texts Related to Positive Energy Districts (PEDs) - General 1

* Required

Instructions.

Please review each text based on the provided question and evaluate it on accuracy, relevance, readability, and scientific quality. Use the 1–5 scale provided for each question, where 1 represents the lowest score and 5 the highest.

Do not compare the quality or content of one answer to another when making your evaluations.

2. **Question:**
What would be the key technologies for PED development in 5 years?

Answer:
Based on the context provided, the key technologies for Positive Energy District (PED) development in the next five years would likely include:

1. **Energy Monitoring and Control Systems**: Sophisticated energy monitoring and control architectures are essential. These systems will likely incorporate increasing degrees of automation and innovation in renewable energy technologies. Communication and control technologies are crucial due to their direct influence on costs and operational efficiency.

2. **Interoperability and Standardization**: Improving interoperability and standardization of systems is critical. These technologies will enhance the digitalization of city, building, and district levels, which is vital for the integration and operation of PEDs.

3. **Energy Flexibility Technologies**: Technologies related to energy flexibility, such as second-life electricity storage devices, will be important. These technologies contribute to more efficient energy use and storage solutions.

4. **Smart Devices and IoT**: The integration of smart meters, sensors, programmable logic controllers (PLCs), and other electronic devices like IoT technology will be necessary. These devices will support the intelligent management of energy resources.

5. **Vehicle-to-Grid (V2G) and Heat Pumps**: Technologies such as vehicle-to-grid systems and heat pumps will play a significant role in energy management and efficiency within PEDs.

6. **Regulatory and Infrastructural Support**: While not a technology per se, the development of supportive regulatory and infrastructural frameworks will be necessary to overcome legal, administrative, and financial barriers, thereby fostering the deployment and integration of these technologies.

Overall, these technologies will collectively enable the efficient and sustainable operation of Positive Energy Districts by enhancing energy management, storage, and distribution capabilities.
*

3. **Question:**
In your opinion, what are the most significant barriers to implementing PEDs in high-density urban environments, and how might these be overcome?

Answer:
opinion significant barriers implementing PEDs high density urban environments might overcome Source several limitations developing pebs positive_energy district gaps con cern need depth research evaluation methodologies design construction use practices transformation regulatory technological infrastructural socio economic context tab Source identification efficient needed policy supporting development positive_energy district recognizing priorities technological social perspectives results proposed paper could suggest urban planners embrace integrated holistic approaches city developments help local administrations develop living laboratories addition fostering social inclusion community members Source many study focus building positive_energy district explore mobility system residents living within district balancing energy requirements design building transportation system public spaces essential optimizing energy_efficiency creating livable environment mobility study literature mobility study generally made charging electrical vehicles
*

4. **Question:**
What are the different targets that PEDs can achieve?

Answer:
Positive Energy Districts (PEDs) can achieve several distinct targets. Firstly, they can promote the generation of energy to yield better results, exporting less and importing different generation potentials. This is largely dependent on the PED's target value. The aim here is to reduce the delivered energy Key Performance Indicator (KPI) and promote higher amounts of site generation.

Secondly, PEDs can also aim for good performance in terms of energy production function and a significant reduction in energy demand. This, in turn, diminishes the self-consumption KPI. However, when accounting for primary export and import, the energy efficiency function may not be properly assessed.

Thirdly, they can focus on social targets characterized by the impact of renewable energy community (REC) actions on the community, reducing energy poverty, improving services, and quality of life.

Lastly, from a technological operation aspect, PEDs can aim to improve the design and operation of RECs, and are linked to three main targets: energy audits, renewable energy, and technological solutions. Other targets can also include carbon dioxide emission reduction, increased installed capacity of renewable energy sources, and increased people's satisfaction.

Please note that these targets can vary based on the specific goals and strategies of the PEDs.
*

5. **Question:**
What strategies are there to include stakeholders at different stages of PED development?

Answer:
Incorporating stakeholders at various stages of Positive Energy District (PED) development is crucial for ensuring community buy-in, relevance, and sustainability of the projects. Here are key strategies for engaging stakeholders throughout the PED development process:

1. Identify and Map Stakeholders

Begin by identifying all potential stakeholders who might be affected by or have an influence on the PED project. This includes local residents, business owners, local government officials, utility companies, property developers, and civic groups. Mapping these stakeholders helps to understand their interests, influence, and potential contributions to the project.

2. Early and Continuous Engagement

Engage stakeholders from the outset of the project to ensure their input is incorporated in the planning phase. Continuous engagement through regular updates and consultations can help maintain their interest and support throughout the project lifecycle.

3. Stakeholder Workshops and Forums

Organize workshops and forums where stakeholders can voice their opinions, provide feedback, and share ideas. This collaborative approach not only promotes transparency but also helps in gathering diverse insights which can enhance the project's design and execution.

4. Surveys and Public Consultations

Conduct surveys and public consultations to collect broader community input. These tools are effective in understanding the community's needs and expectations, which can guide the planning and implementation phases of PED projects.

5. Co-Creation Sessions

Implement co-creation sessions where stakeholders can actively participate in creating solutions or elements of the PED. This empowers stakeholders and ensures that the project aligns with local needs and values.

6. Advisory Committees

Form advisory committees consisting of representatives from different stakeholder groups. These committees can provide ongoing advice and review project progress, acting as a bridge between the project team and the broader community.

7. Transparent Communication

Maintain a transparent communication strategy that keeps stakeholders informed about project developments, challenges, and successes. Utilize various communication channels such as newsletters, websites, social media, and public notices to reach different audience segments.

8. Pilot Projects

Implement pilot projects as a practical method of demonstrating PED concepts. Pilots allow stakeholders to experience the benefits firsthand, provide feedback, and understand the project's potential impact on their community.

9. Educational Programs

Develop educational programs and materials to inform stakeholders about the benefits and workings of PEDs. Educated stakeholders are more likely to support and participate actively in the project.

By employing these strategies, PED projects can effectively include stakeholders in the development process, ensuring that the projects are well-supported and successfully integrated into the community's fabric.

6. **Question:**
What distinguishes net-zero energy buildings from positive energy districts?

Answer:
positive_energy_district concept developed previous ideas nearly net positive zero energy building extending single building district scale order exploit energy mutuality building urban decarbonization need broaden perspective arises several key aspects improvements single building approach example district approaches could offer benefits terms energy_system design integration terms striving towards optimized urban morphology district scale energy_system could offer economical technical scale Source net zero positive energy building literature net pebs conventionally described building produce energy consume place analogy nzebs consider guided principles although nzebs npebs share many characteristics many differences precisely within energy trading network calculation period systemic assessment performance
*

Feedback about PED texts, part 2 of 2

You are half-way done!
Please review each text based on the provided question and evaluate it on accuracy, relevance, readability, and scientific quality. Use the 1–5 scale provided for each question, where 1 represents the lowest score and 5 the highest.

Do not compare the quality or content of one answer to another when making your evaluations.

7. **Question:**
In your opinion, what are the most significant barriers to implementing PEDs in high-density urban environments, and how might these be overcome?

Answer:
Based on the given context, the significant barriers to implementing Positive Energy Districts (PEDs) in high-density urban environments include the need for in-depth research and evaluation methodologies for design, construction, and use practices. These methodologies should be tailored to the specific characteristics and needs of the district, as one-size-fits-all solutions may not be effective. Technological, infrastructural, regulatory, and socio-economic contexts also play a key role in the implementation of PEDs.

Other barriers include complexities related to the design and implementation of energy-efficient mobility systems, as well as the allocation of energy and emission provisions for public transport. The occupancy density and the allocation of trips and floor usage within the district can also present challenges.

Overcoming these barriers may involve several strategies. Firstly, the development of efficient policies to support the development of PEDs is crucial. Recognizing and addressing priorities from both technological and social perspectives is also important. Secondly, urban planners could embrace integrated and holistic approaches to city development, which can aid local administrations in developing living districts that foster social inclusion.

Furthermore, to overcome the mobility-related challenges, research should be focused on balancing the energy requirements of the design and building of transportation systems and public spaces. This could involve exploring the use of electric vehicles and other sustainable mobility solutions. Lastly, the methodologies for the design of PEDs should be tailored to the unique characteristics of each district, enhancing the efficiency and performance of these districts.
*

8. **Question:**
What distinguishes net-zero energy buildings from positive energy districts?

Answer:
current regulation practices already leading near zero energy building baseline however positive _ energy _ district energy _ balance may include energy _ use appliances sources directly connected building achieving benchmark even challenging existing building particularly renovation process positive _ energy _ district concept developed previous ideas nearly net positive zero energy building extending single building district scale order exploit energy mutuality building urban decarbonization need broaden perspective arises several key aspects improvements single building approach example district approaches could offer benefits terms energy _ system design integration terms striving towards optimized urban morphology district scale energy _ system could offer economical technical scale net zero positive energy building literature net pebs conventionally described building produce energy consume place analogy nzebs consider guided principles although nzebs npebs share many characteristics many differences precisely within energy trading network calculation period systemic assessment performance
*

9. **Question:**
What strategies are there to include stakeholders at different stages of PED development?

Answer:
The strategies to include stakeholders at different stages of Positive Energy District (PED) development involve several key approaches:

1. **Characterization and Identification**: Initially, it is crucial to characterize and identify stakeholders effectively. This involves recognizing individuals and groups with vested interests in sustainable development, such as inhabitants and local municipalities, as well as less obvious stakeholders whose actions impact or are affected by the PED's formulation and implementation.

2. **Structured Framework for Collaboration**: Although structured frameworks may be absent in some contexts, the literature suggests that activities aimed at supporting cooperation among stakeholders are typically included in PED design methodologies. This implies the need for a structured framework to facilitate collaboration and exchange of initiatives among stakeholders, especially in developing countries.

3. **Multidisciplinary Collaboration**: Effective multidisciplinary collaboration among stakeholders is paramount throughout the design phases of PED development. This includes engaging various disciplines and sectors to ensure a comprehensive approach to PED design and implementation.

4. **Spatial-Tiered Stakeholder Categorization**: A spatial-tiered approach to stakeholder categorization is advocated to emphasize the importance of multi-level decision-making. This involves categorizing stakeholders based on spatial and administrative levels to ensure their interests and needs are considered at different stages of the PED process.

5. **Flexible Frameworks for Customization**: The PED framework should be flexible enough to allow for national adaptations and customization. This flexibility is necessary to tailor the PED development process to the specificities of different urban areas and the diverse needs of stakeholders involved.

6. **Phased Approach**: The PED development process is divided into key phases, including master planning, energy planning, construction and implementation, and monitoring. Each phase requires the alignment of goals and tools with stakeholder interests, ensuring their active involvement and participation throughout the process.

7. **Public Engagement and Transformation Guidance**: Engaging public authorities and stakeholders is essential for transforming existing districts into PEDs. This involves guiding stakeholders through the transformation process and addressing technical, administrative, social, and economic topics to overcome planning and implementation pitfalls.

8. **Roadmaps and Practical Tools**: Providing tailored practical roadmaps and tools to managers, local authorities, and stakeholders, including roles for consumers and producers, is critical. This aids in facilitating the PED transition and ensuring stakeholders are effectively integrated into the process.

By applying these strategies, stakeholders can be effectively included at different stages of PED development, ensuring their diverse interests and priorities are addressed, which is crucial for the successful implementation of sustainable urban projects.
*

10. **Question:**
What are the different targets that PEDs can achieve?

Answer:
? generation promoted
*

11. **Question:**
What would be the key technologies for PED development in 5 years?

Answer:
Based on the context provided, the key technologies for Positive Energy District (PED) development in the next five years could include sophisticated energy monitoring and control architecture, renewable energy technology, communication and control technology, and various hardware and software solutions. More specifically, these technologies can be detailed as follows:

1. Energy Monitoring and Control Architecture: This includes automation and innovation in life cycle management. It is important for realizing energy communities and managing operations.

2. Renewable Energy Technology: The degree of maturity of renewable energy technology would be a determining factor. This technology could include solutions for both generation (like solar panels, wind turbines) and storage (like batteries).

3. Communication and Control Technology: These technologies have a direct influence on costs and are relevant to the technological aspects of PEDs. They may include advanced grid communication systems, smart metering, and other IoT technologies.

4. Hardware and Software Solutions: The context mentions the use of second-life electricity storage devices for energy flexibility, long-term and short-term monitoring systems, sensors, smart meters, Programmable Logic Controllers (PLCs), and energy management systems. The integration of electronic devices like IoT technology is also emphasized, which contributes to the digitalization of city and district levels.

5. Interoperability and Standardization: Improving interoperability and standardization of the system could boost the digitalization of city and district levels, making it a key aspect of future PED development.

6. Legal and Administrative Support: The removal of legal, administrative, and financial barriers is also crucial to allow early stage technology to reach market maturity. This support could be important for promoting the development and growth of PEDs.

It's important to note that the exact technologies and their relevance may vary based on advancements in technology, regulatory changes, and specific geographical or local conditions.
*

	1	2	3	4	5
How accurate is the provided answer? (1 = Completely inaccurate, 5 = Completely accurate)	○	○	○	○	○
How well does the provided answer address the question? (1 = Not at all relevant, 5 = Highly relevant)	○	○	○	○	○
How easy is the provided answer to read and understand? (1 = Very difficult, 5 = Very easy)	○	○	○	○	○
How scientifically rigorous is the provided answer (e.g., logical reasoning, structured argumentation)? (1 = Poor, 5 = Excellent)	○	○	○	○	○
Does the provided answer avoid unnecessary repetition and verbosity? (1 = Very verbose, 5 = Very concise)	○	○	○	○	○
How confident do you feel about the reliability of this answer? (1 = Not confident, 5 = Very confident)	○	○	○	○	○

Appendix B: Model outputs for a factual question:

"What renewable energy sources were considered in the case study of the Riga Technical University campus?"

BERT: modelled system includes building owned riga technical university rtu see figure table located peninsula kipsala riga latvia include research building dormitories swimming pool detailed information enclosing structures engineering system gathered building built recently renovated within last decade research building still building phase however potential energy _ demand estimated building parameters several building data centres server rooms located considered potential heat source hp installations indirectly transition fossil fuel based energy _ system renewable energy _ source become crucial consideration national scale planning processes

BERT-GPT4o: The case study of the Riga Technical University campus considered several renewable energy sources. These included solar photovoltaic (PV) panels, both vertical and horizontal wind turbines, and heat recovery from data centers' cooling units. Additionally, the study explored the integration of air-to-water and air heat pumps (HPs) as part of the renewable energy systems. These renewable energy sources were part of the effort to transition away from fossil fuels and to establish a decarbonized, positive energy district on the campus.

ChatGPT: In the case study of the Riga Technical University (RTU) campus, several renewable energy sources were considered to transition towards a Positive Energy District (PED):

1. **Solar Photovoltaic (PV) Panels**: The study evaluated the installation of building-integrated solar panels, analyzing configurations and orientations to optimize energy generation profiles.

Ortus

2. **Wind Energy**: The potential for installing wind generators around the RTU campus was discussed as a means to harness wind energy.

Ortus

3. **Hydropower**: The study explored the integration of wind energy storage with existing hydropower plant reservoirs in Latvia, utilizing pumped hydroelectric energy storage (PHES) technology to accommodate the intermittency of wind electricity generation.

Ortus

4. **Heat Recovery and Waste Heat Integration**: The potential for heat recovery and waste heat integration from data centers and cooling systems was evaluated to cover the campus's heat demand.

Ortus

5. **Water Source Heat Pumps**: The study considered the use of water source heat pumps as part of the renewable energy solutions for the campus.

Ortus

6. **Off-Site Wind Turbines**: The integration of off-site wind turbines was analyzed to supplement the campus's renewable energy portfolio.

Ortus

These renewable energy sources were assessed in combination with energy storage solutions, such as thermal storage, batteries, hydrogen, and fuel cells, to enhance the campus's energy efficiency and sustainability.

Ortus

Sources

NFT T5GPT: The case study of the Riga Technical University campus considered multiple renewable energy sources. Specifically, the study looked into the implementation of a smart energy system with multiple energy sources and energy efficiency measures. These included power generation through building-integrated solar panels as per the abstract of the document. In addition, the introduction of the document mentions the exploration of various renewable energy system technology such as solar PV panels and vertical and horizontal wind turbines. The research also considered heat recovery from data centres and cooling units as potential energy sources. The conclusion section further mentions the use of pioneering innovative technology for hydrogen production.

T5: renewable_energy sources considered case_study Riga Technical University campus source modelled system includes building owned riga technical university rtu see figure table located peninsula psala riga latvia include research building dormitories swimming pool detailed information enclosing structures engineering system gathered building recently renovated within last decade research building still building phase however potential energy_demand estimated building parameters several building data centres server rooms located considered potential heat source hp installations indirectly Source transition fossil fuel based energy_system renewable_energy_source become crucial consideration national scale planning processes local scale energy_system planning research conducted seek technical solution specific district

T5-GPT4: The case study of the Riga Technical University campus considered several renewable energy sources for the transition towards a positive energy district. These sources include building-integrated solar panels and various renewable energy system (RES) technologies such as solar PV panels and heat recovery from data centres and cooling units. The study also explored the use of vertical and horizontal wind turbines. Furthermore, the case study considered the innovative technology of hydrogen production for energy.

Appendix C: Model outputs for an opinion question:

How might the Cities4PEDs project, involving Brussels, Vienna, and Stockholm, influence the development of PEDs in smaller European cities?

BERT: back possibility access funding needed could initiated less three years city important projects aligned strategy agendas demonstrate results broad european project could city list selected project ideas two per city provided table final list project ideas developed details implementation plans conference smart sustainable planning city regions sspcr bolzano december positive _ energy _ district definition development v4 brussels meeting reflections horizon europe mission area climate neutral smart _ city included programme city obviously considered upscaling pebs positive _ energy _ district towards pecs pers positive energy city regions currently established terms steps towards eu goal climate neutrality strategic alignment city level scaling ambitions present early example cityxchange definition work city substantially contribute european _ green _ deal targets accelerating transition towards smarter efficient renewable based energy models context european city need increase self sufficiency resilience energy _ system especially promotion district neighbourhood scale energy projects

BERT-GPT4o: The Cities4PEDs project, which involves major cities like Brussels, Vienna, and Stockholm, can significantly influence the development of Positive Energy Districts (PEDs) in smaller European cities through several mechanisms:

1. **Knowledge Sharing and Best Practices**: The project can serve as a model for smaller cities by sharing insights and best practices developed in these larger cities. The experiences and methodologies applied in Brussels, Vienna, and Stockholm can provide smaller cities with a framework for implementing their own PEDs. This includes strategies for energy planning, governance measures, and stakeholder engagement.

2. **Scaling and Adaptation**: Smaller cities can learn from the scaling ambitions and strategic alignments that are being tested in the Cities4PEDs project. By adapting successful strategies from larger cities, smaller cities can more efficiently implement PEDs tailored to their specific contexts and needs.

3. **Capacity Building and Skill Development**: The project can help address the lack of skilled personnel in smaller cities by offering training and capacity-building initiatives. This can empower local authorities and stakeholders in smaller cities to effectively plan and manage PED projects.

4. **Promotion of a Portfolio Approach**: The Cities4PEDs project emphasizes a portfolio approach to address a range of contexts and challenges. Smaller cities can adopt this approach, which allows for flexibility and adaptability in implementing PEDs across different urban areas and conditions.

5. **Policy and Governance Frameworks**: By observing the governance frameworks and policy measures implemented in the Cities4PEDs project, smaller cities can develop their own supportive policies and regulations that encourage the development of PEDs.

6. **Increased Access to Funding and Resources**: Participation in a broader European initiative like Cities4PEDs can open up opportunities for smaller cities to access funding and resources that are crucial for the implementation of PEDs.

Overall, the Cities4PEDs project can act as a catalyst for smaller European cities to embark on their own journeys towards climate neutrality and energy resilience by leveraging the experiences and innovations from Brussels, Vienna, and Stockholm.

ChatGPT: The **Cities4PEDs** project, a collaboration among Brussels, Vienna, and Stockholm, aims to explore and facilitate the development of Positive Energy Districts (PEDs) across diverse urban contexts. The insights and methodologies derived from this initiative can significantly influence the adoption of PEDs in smaller European cities in several ways:

1. Development of Practical Tools and Resources

The project has produced comprehensive resources, such as the **PED Atlas** and the **Working Paper: Enabling PEDs through city instruments**, which offer detailed analyses and strategies for implementing PEDs. These documents provide smaller cities with actionable insights and frameworks that can be tailored to their specific contexts, enabling them to embark on PED projects with greater confidence and clarity.

Energy Cities

2. Establishment of a Transnational Community of Practice

By fostering a network of cities and stakeholders engaged in PED development, Cities4PEDs facilitates knowledge exchange and capacity building. Smaller cities can tap into this community to learn from the experiences of larger counterparts, access best practices, and receive mentorship, thereby accelerating their own PED initiatives.

JPI Urban Europe

3. Addressing Legal and Institutional Challenges

The project delves into the legal, institutional, and organizational adjustments necessary for PED realization. Smaller cities can benefit from these findings by understanding potential regulatory hurdles and adopting recommended solutions, streamlining their path toward sustainable energy districts.

Energy Cities

4. Emphasis on Citizen Engagement

Recognizing the importance of community involvement, Cities4PEDs explores methods for effective citizen participation in PED projects. Smaller cities, often characterized by close-knit communities, can leverage these strategies to foster local support and co-ownership of energy initiatives, enhancing the success and sustainability of PEDs.

Energy Cities

5. Adaptation to Diverse Urban Contexts

The project's focus on diverse urban settings ensures that the PED concepts and tools developed are versatile and adaptable. Smaller cities can modify these approaches to align with their unique socio-economic and spatial characteristics, ensuring that PED implementation is contextually relevant and effective.

JPI Urban Europe

In summary, the Cities4PEDs project provides a robust framework and valuable resources that smaller European cities can utilize to develop Positive Energy Districts, fostering sustainable urban transformation across the continent.

Sources

NFT T5GPT: The Cities4PEDs project, involving Brussels, Vienna, and Stockholm, could influence the development of Positive Energy Districts (PEDs) in smaller European cities in several ways. Firstly, these cities can serve as role models demonstrating successful implementation of PEDs. The strategies, methodologies, and experiences from these cities can be replicated in smaller cities, as seen in the implementation of 10 replicated solutions in Maia, Reykjavik, Kifissia, Kladno, and Lviv.

Secondly, these cities can contribute to the development of a more comprehensive and practical PED definition, addressing the practical implementation challenges from a bottom-up perspective. Such a definition could aid smaller cities in understanding and adopting the concept of PEDs.

Thirdly, the project could foster an energy planning culture and promote the training of adequately skilled personnel in smaller cities. This might be achieved through sharing best practices, knowledge transfer, and providing training based on the experiences of the larger cities.

Finally, the Cities4PEDs project may facilitate strategic alignment at city level and scaling ambitions, which are crucial for smaller cities aiming to contribute to the European Green Deal targets and increase their self-sufficiency and resilience in their energy systems. This can be seen in the example of Stockholm's Royal Seaport district redevelopment project, which adopted an interscalar approach based on interaction and continuous consultation with stakeholders.

However, the exact influence of the Cities4PEDs project on smaller European cities would depend on specific factors such as the cities' existing infrastructure, resources, and policies, as well as their willingness and capacity to adopt and adapt the PED concept and practices.

T5: how might Cities4PEDs project involving Brussels Vienna Stockholm influence development PEDs smaller European city Source back possibility access funding needed could initiated less three years city important projects aligned strategy agendas demonstrate results broad european project could city list selected project ideas two per city provided table final list project ideas developed details implementation plans Source conference smart sustainable planning city regions sspcr bolzano december positive_energy_district definition development v4 brussels meeting reflections horizon europe mission area climate neutral smart_city included programme city obviously considered upscaling pebs positive_energy_district towards

T5-GPT4: The Cities4PEDs project, involving Brussels, Vienna, and Stockholm, could influence the development of Positive Energy Districts (PEDs) in smaller European cities in several ways, based on the context provided.

First, these cities could serve as role models demonstrating how to align city-level projects and strategies with broader European agendas. As indicated in the first document, cities can initiate important projects that align with strategy agendas to demonstrate results for a broad European project. This could inspire smaller cities to initiate similar projects and strategies.

Second, the Cities4PEDs project could provide practical experiences and lessons that could be fed back into the European PED framework definition development process. As mentioned in the second document, the development of the PED definition has included reflections from these cities, which were considered for upscaling PEDs towards Positive Energy City Regions (PECRs). This could help smaller cities understand the practical implementation challenges and potential solutions in developing PEDs.

Third, the project could help smaller cities increase their self-sufficiency and resilience in their energy systems. The third document suggests that cities need to transition towards smarter, efficient, renewable-based energy models, especially by promoting district and neighborhood scale energy projects such as PEDs.

Fourth, the Cities4PEDs project could encourage a portfolio approach in smaller cities. The fourth document mentions that complex city projects could benefit from a portfolio approach that addresses a range of contexts linked to EU city missions.

Lastly, the Cities4PEDs project could influence the development of PEDs in smaller European cities by showcasing innovative approaches to urban regeneration. The fifth document highlights the interscalar approach adopted in Stockholm's Royal Seaport District redevelopment, which focuses on governance measures based on interaction and continuous consultation with involved stakeholders such as the municipality, investors, and inhabitants. This could provide a model for smaller cities on how to effectively engage stakeholders in PED projects.

Appendix D: Model outputs for a synthetic question:

What are specific examples of how PEDs are implemented in Borlänge?

BERT: reference relevant information collected allow better de scription pebs positive _ energy _ district favour definition depth

BERT-GPT4o: The context provided does not include specific examples of how Positive Energy Districts (PEDs) are implemented in Borlänge. The text primarily discusses general aspects of PEDs, such as the importance of defining boundaries for system assessment and the use of a life cycle assessment (LCA) approach. However, there are no detailed descriptions or examples related to the specific implementation of PEDs in Borlänge.

ChatGPT: In **Borlänge**, Sweden, the implementation of **Positive Energy Districts (PEDs)** is exemplified by the **Jakobsdalen urban development project**. This initiative, managed by Borlänge Kommun Förvaltning AB, focuses on sustainable community development through:

- **New Housing and Businesses**: Integrating residential and commercial spaces to foster a vibrant community.
- **Urban Gardening**: Promoting local food production and green spaces.
- **Sustainable Traffic Solutions**: Enhancing mobility with eco-friendly transportation options.

Dalarna University has contributed expertise to this project, facilitating collaboration across sectors and providing a platform for research and innovation in areas like smart transport and energy efficiency. Jakobsdalen serves as a testbed for these concepts, offering concrete examples of sustainable development that can be replicated in other regions.

Dalarna University

Additionally, Dalarna University offers a course titled **Positive Energy Districts**, aimed at equipping students with the skills to design, plan, and simulate PEDs. This educational effort supports the broader goal of implementing PEDs in Borlänge by fostering a knowledgeable workforce capable of advancing such initiatives.

Dalarna University

Through projects like Jakobsdalen and dedicated educational programs, Borlänge is actively pursuing the development of Positive Energy Districts, integrating sustainable practices into urban planning and community engagement.

Sources

NFT T5GPT: The provided context does not include any specific information about the implementation of Positive Energy Districts (PEDs) in Borlänge. Therefore, it's not possible to provide specific examples or details based on the given context.

T5: positive_energy_district favour definition depth state energy positive paradigm fig analysis topics covered suggests research pebs positive_energy_district deepens understanding experiences concerning implementation principles especially according lca approach rare subsequently scope assessment defined respective articles analysed fig delimiting calculation methodology boundaries system assessment boundary used various study manifold Source renewable_energy photovoltaics heat pumps mobility solution like electric vehicles however explained section resource pool positive_energy_district encompasses technology base tangible intangible resources knowledge exchange citizens engagement furthermore several endogenous variables clds reflect various key performance indicators enable assessment public goals carbon dioxide emission reduction installed capacity etc internal goals increased people satisfaction selected kpis see supplementary table

TF-GPT4: The provided context does not give any specific examples or information about how Positive Energy Districts (PEDs) are implemented in Borlänge. Therefore, it's not possible to provide a detailed, specific, and comprehensive answer based on the given context.

Appendix E: 40 Questions Extracted from Academic Articles

1. What is the role of tools and methods in positive energy districts design?
2. How do PEDs embrace data-driven design?
3. What is the purpose of multi-stakeholder engagement tools?
4. What are the differences between net zero energy buildings and positive energy districts?
5. How different scales and evaluation boundaries are changing the analytical perspective of PEDs?
6. What is involved in the planning of PEDs?
7. Which metrics, benchmarks and standards are used to assess PEDs?
8. What are the available analytical approaches and methods to evaluate PED performance
9. How are physic based models used to simulate energy use in PEDs?
10. How can digital workflows be leveraged to achieve advanced PED designs?
11. Which new PED tools are needed considering future climate and urban resilience?
12. How are PED tools integrated into the design phases and with which stakeholders do they interact?
13. What are the gaps between the existing tools and PED design in practice
14. What are the outlooks, future challenges, and opportunities for PED tools?
15. What are the opportunities of transitioning from individual buildings to districts?
16. What are the challenges of transitioning from individual buildings to districts?
17. What are the challenges of multi-objective environmental analysis?
18. What are the opportunities of multi-objective environmental analysis?
19. What are the opportunities with multi-stakeholder engagement?
20. What are the challenges with multi-stakeholder engagement?
21. What are the opportunities with data-driven design in PEDs?
22. What are the challenges with data-driven design in PEDs?
23. What enabling conditions are required to support rapid scaling up of PEDs in Europe?
24. What are framework conditions within PEDs?
25. What is Prefiguration in relation to PEDs?
26. What are the emerging impacts of PEDs?
27. What relational components are essential for a city to successfully implement PEDs?
28. Which structural aspects are key for the effective implementation of PEDs?
29. What is an urban governance framework for PEDs
30. What engagement strategies in PED implementation can ensure fruitful co-creation processes?
31. What collaborative governance processes and functionalities undergird PEDs?
32. What would prototype design patterns for PEDs look like?
33. How can diverse stakeholders create a critical mass to implement PED?
34. How does a PED get started?
35. What is the motivation for starting a PED
36. In what ways can PEDs advance equitable economic development, i.e. socio-economic sustainability?
37. In what ways can PEDs contribute to reductions in energy demand, i.e. ecological sustainability?
38. How can PED's lead to developing the energy system?
39. What does the implementation of PEDs imply for urban futures?
40. What lessons can we learn from PEDs?

Appendix F: Sample Retrieved Document Sections with Similarity Scores

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Query: What would be the key technologies for PED development in 5 years?
Retrieved Section 1 | Similarity Score: 0.51
Source: A-Comprehensive-PEDDatabase-for-Mapping-and-Comparing-Positive-Energy-Districts-Experiences-at-European-Level_2022_MDPI.pdf.xml
Section Title: • | Text (first 100 chars): technological solution etc...

Retrieved Section 2 | Similarity Score: 0.49
Source: Renewable-Energy-Communities-in-Positive-Energy-Districts-A-Governance-and-Realisation-Framework-in-Compliance-with-the-Italian-Regulation_2023_MDPI.pdf.xml
Section Title: Analysis of success and critical factors | Text (first 100 chars): respectively tend towards contraction granting long term loans realising investments required build ...

Retrieved Section 3 | Similarity Score: 0.48
Source: Positive-Energy-Buildings-and-Districts-beyond-the-NZEB-paradigm-towards-a-wholelife-approach_2023_Firenze-University-Press.pdf.xml
Section Title: Research gaps and application barriers | Text (first 100 chars): several limitations developing pebs positive_energy_district gaps con cern need depth research evalu...

Retrieved Section 4 | Similarity Score: 0.47
Source: Network-dynamics-of-positive-energy-districts-a-coevolutionary-business-ecosystem-analysis_2023_Frontiers-Media-SA.pdf.xml
Section Title: "so, we created an energy community and we are operating it to understand and investigate what works, what does not, what are the technical limits, what are the opportuni

Retrieved Section 5 | Similarity Score: 0.47
Source: Positive-Energy-Districts-Fundamentals-Assessment-Methodologies-Modeling-and-Research-Gaps_2024_Multidisciplinary-Digital-Publishing-Institute-MDPI.pdf.xml
Section Title: Technologies in positive energy districts: development, use and barriers | Text (first 100 chars): approach second life electricity_storage devices terms energy_flexibili

Tokens used: 919
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Query: What renewable energy sources were considered in the case study of the Riga Technical University campus?
Retrieved Section 1 | Similarity Score: 0.57
Source: Transition-towards-Positive-Energy-District-A-Case-Study-from-Latvia.pdf.xml
Section Title: Framework of analysed district | Text (first 100 chars): modelled system includes building owned riga technical university rtu see figure table located penin...

Retrieved Section 2 | Similarity Score: 0.55
Source: Transition-towards-Positive-Energy-District-A-Case-Study-from-Latvia.pdf.xml
Section Title: Abstract | Text (first 100 chars): transition fossil fuel based energy_system renewable energy_source become crucial consideration nati...

Retrieved Section 3 | Similarity Score: 0.54
Source: Transition-towards-Positive-Energy-District-A-Case-Study-from-Latvia.pdf.xml
Section Title: Introduction | Text (first 100 chars): primary research question cost optimal solution decarbonizing university campus district integrating...

Retrieved Section 4 | Similarity Score: 0.52
Source: Design-Modelling-and-Performance-Evaluation-of-a-Positive-Energy-District-in-a-Danish-Island_2022_Ubiquity-Press.pdf.xml
Section Title: Introduction | Text (first 100 chars): renewable_energy system capacity become fossil fuel free study provide preliminary results forming b...

Retrieved Section 5 | Similarity Score: 0.51
Source: Transition-towards-Positive-Energy-District-A-Case-Study-from-Latvia.pdf.xml
Section Title: Conclusions and discussion | Text (first 100 chars): allow increased flexibility allow choose beneficial power heating sources depending external conditi...

Tokens used: 706
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Query: Why is Vallastaden district in Linköping, Sweden, suitable for being a PED?
Retrieved Section 1 | Similarity Score: 0.57
Source: Towards a positive energy balance - A comparative analysis of the planning and design of four positive energy districts and neighbourhoods in Norway and Sweden.pdf.xml
Section Title: Vallastaden (swe) | Text (first 100 chars): Vallastaden (see Fig.  5 ) is an urban development project, located 3.5
km from the city centre of L...

Retrieved Section 2 | Similarity Score: 0.57
Source: Towards a positive energy balance - A comparative analysis of the planning and design of four positive energy districts and neighbourhoods in Norway and Sweden.pdf.xml
Section Title: Process | Text (first 100 chars): PEDs can be developed through different management structures, and Verksbyen falls in the category w...

Retrieved Section 3 | Similarity Score: 0.54
Source: Towards a positive energy balance - A comparative analysis of the planning and design of four positive energy districts and neighbourhoods in Norway and Sweden.pdf.xml
Section Title: Scope and limitations | Text (first 100 chars): This paper investigates the planning and design processes of PEDs and SPENs by conducting a comparat...

Retrieved Section 4 | Similarity Score: 0.53
Source: An-Exploratory-Study-on-Swedish-Stakeholders-Experiences-with-Positive-Energy-Districts_2023_MDPI.pdf.xml
Section Title: Conclusions | Text (first 100 chars): PED is a recent concept and there is ambiguity on many facets of PEDs. Harnessing experiences is vit...

Retrieved Section 5 | Similarity Score: 0.52
Source: Network-dynamics-of-positive-energy-districts-a-coevolutionary-business-ecosystem-analysis_2023_Frontiers-Media-SA.pdf.xml
Section Title: Public value proposition | Text (first 100 chars): The PED concept is seen as a vision to guide local renewable energy transition. The public value pro...

Tokens used: 2259
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Query: What is the primary reason that PEDs in Säter primarily utilize wind power?
Retrieved Section 1 | Similarity Score: 0.49
Source: Assessment of energy systems configurations in mixed-use Positive Energy Districts through novel indicators for energy and environmental analysis.pdf.xml
Section Title: Scenarios without hydrogen supply | Text (first 100 chars): pronounced lower producibility wind turbines comparison sc opposite results obtained comparison sc f...

Retrieved Section 2 | Similarity Score: 0.46
Source: A novel methodology and a tool for supporting the transition of districts and communities in Positive Energy Districts.pdf.xml
Section Title: Validation, replicability and upscaling | Text (first 100 chars): determined assuming efficiency pg equal fig shows preliminary results obtained sc installation kw wi...

Retrieved Section 3 | Similarity Score: 0.46
Source: Energy-Management-System-for-a-Residential-Positive-Energy-District-Based-on-Fuzzy-Logic-Approach-RESTORATIVE_2024_Multidisciplinary-Digital-Publishing-Institute-MDPI.pdf.xml
Section Title: Scalability of restorative | Text (first 100 chars): incorporate wind turbines renewable technology flexibility ensures system tailored leverage abundant...

Retrieved Section 4 | Similarity Score: 0.46
Source: Recommendations for a positive energy district framework – Application and evaluation of different energetic assessment methodologies.pdf.xml
Section Title: Zero emission neighbourhood | Text (first 100 chars): total local energy_generation sc total energy_use sg greater urban district however discon variant y...

Retrieved Section 5 | Similarity Score: 0.46
Source: Assessment of energy systems configurations in mixed-use Positive Energy Districts through novel indicators for energy and environmental analysis.pdf.xml
Section Title: Scenarios without hydrogen supply | Text (first 100 chars): preliminary insights supply renewable electricity min timestep res el 0 shown fig switch sc determin...

Tokens used: 740
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